# REAPER-Notator

## Complete User Manual & Functional Reference

*Native Score Editing, Engraving & DAW Automation for Cockos REAPER*

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| **Version:** | v3.0 (Production Release) |
| **Release Date:** | November 04, 2026 |
| **Compatibility:** | Cockos REAPER v7.0+ (Windows, macOS, Linux) with ReaImGui & Reaticulate |
| **License:** | Proprietary (All Rights Reserved - Copyright (C) 2026 A-intheCode) |

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# Chapter 1: Introduction & Getting Started

## 1.1 What Is REAPER-Notator & What Do You Need?

REAPER-Notator turns Cockos REAPER into a complete, easy-to-use sheet music editor. Instead of exporting your music to a separate notation program and losing your virtual instruments and mixes, you can read, write, record, and arrange sheet music right inside REAPER. Every note, dynamic mark, slur, and articulation you place on the page immediately plays back through your loaded instruments.

What You Need Installed Before Starting:

- Cockos REAPER v7.0 or newer.
- ReaImGui extension (version 0.9 or newer, easily installed through ReaPack).
- Reaticulate (Required so REAPER-Notator can switch your instrument playing techniques like Staccato, Legato, and Pizzicato automatically; free via ReaPack: https://reaticulate.com/).
- SWS / S&M Extension & JS_ReaScriptAPI (recommended, also available in ReaPack).

## 1.2 Tour of the Three-Part Screen Layout

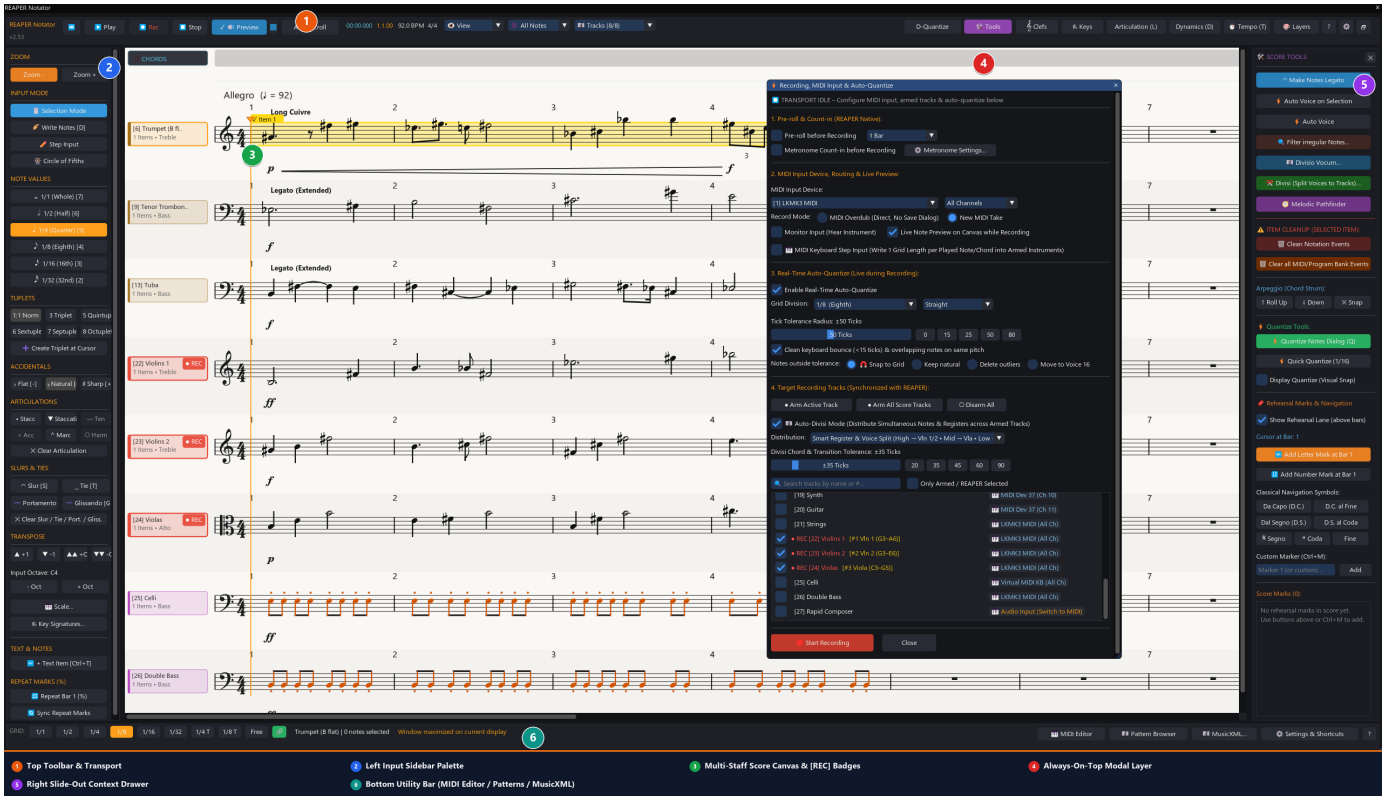



*Figure 1.1: REAPER-Notator 3-Column Workspace - Top Toolbar & Transport (1), Left Input Sidebar (2), Multi-Staff Score Canvas & [REC] Track Headers (3), Always-On-Top Modal Layer (4), Right Context Drawer (5), and Bottom Utility Bar (6).*




*Figure 1.2: Score System Header - Collapsible Chords Lane, Measure Numbers, Tempo Marking (Allegro), Articulation Labels, and Tuplet Brackets.*

The window is divided into three simple areas so everything is within easy reach:

- Left Sidebar (Your Note Toolbox): Pick your mouse mode (Select, Draw, Text, or Erase), choose note lengths (whole notes down to 32nd notes), add dots, sharps, flats, tuplets, slurs, ties, and quick articulations.
- Center Score Area (Your Sheet Music): Your interactive musical page where you see all your instrument staves, measures, notes, chords, lyrics, dynamics, and rehearsal letters.
- Right Side Panels (Slide-Out Drawers): Click any button on the right side of the top toolbar to slide out specialized palettes for Dynamics, Articulations, Tempo, Clefs, Key & Time Signatures, Patterns, Lyrics, Freehand Annotations, and Settings.

## 1.3 Moving, Docking & Maximizing the Window

You can use REAPER-Notator as a free-floating window on any monitor or dock it directly inside REAPER. Use the Zoom controls in the top-left corner (from 60% to 200%) to make the staves as large or compact as you like.

Helpful Window Features:

- Clean Centered Startup: Every time you launch REAPER-Notator, it opens neatly centered on your screen so it never gets lost off-screen.
- Full-Screen Maximize Button (Top-Right or F11 Key): Click the Maximize button in the top-right corner (or press F11) to fill your entire monitor with sheet music. Click it again (or press F11) to return to a centered floating window.
- Works on 4K & Multi-Monitor Setups: You can freely drag the window between a 4K laptop screen and a regular external monitor, and it will scale cleanly without blurry text or cut-off edges.
- Pop-Up Dialogs Stay in Front: Whenever you open a pop-up window-such as Recording Settings, Melodic Pathfinder, Circle of Fifths, or Settings-it stays cleanly on top of the music until you close it.

# Chapter 2: Top Toolbar, Live Recording, Visibility Options & Navigation

## 2.1 Title & Version Display

In the upper-left corner you will see 'REAPER Notator' along with your current version number (v3.0), so you can always tell at a glance which version you are using.

## 2.2 Play, Stop & Record Buttons (Transport)

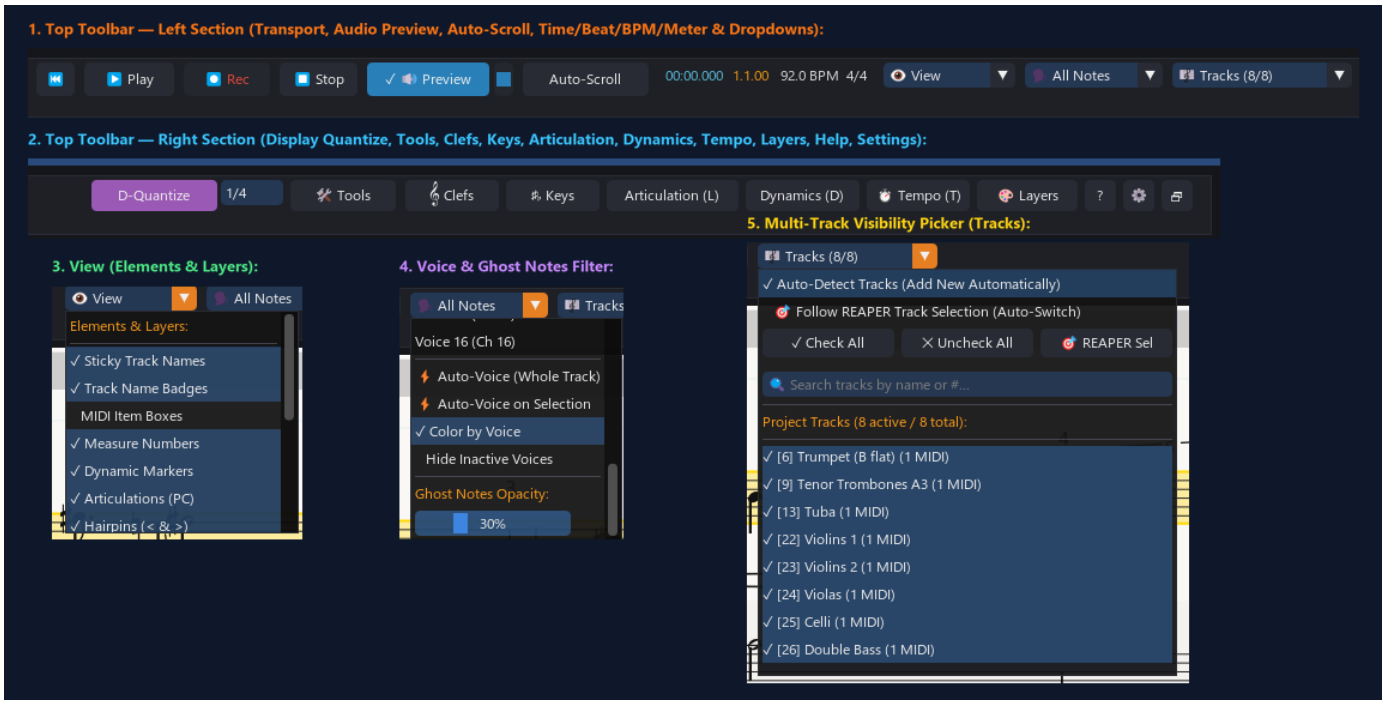



*Figure 2.1: Complete Top Toolbar & Popover Menus - Transport & Readouts (1), Display Quantize & Drawer Toggles (2), View Elements & Layers Filter (3), Voice & Ghost Notes Filter (4), and Multi-Track Visibility Picker (5).*

These buttons control REAPER's playback directly from the score window:

- Rewind (|<): Jumps straight back to the beginning of your project (Bar 1).
- Play / Pause (> / ||): Starts or pauses playback at the current cursor position. Keyboard shortcut: Spacebar.
- Record ([Rec] / [REC]): Located right between Play and Stop.
- Left-Click: Starts recording immediately (the button glows bright red). Click the red button again to stop recording.
- Right-Click: Opens the Recording & Auto-Quantize Settings window, where you can arm tracks, set up a count-in, or turn on MIDI keyboard step input.
- Stop: Stops playback or recording and neatly finalizes any notes you just recorded.

## 2.3 Watching Your Notes Appear Live While Recording

When you press Record and play on your MIDI keyboard, REAPER-Notator shows your performance as sheet music in real time:

- Bright Red Recording Line: A vivid red vertical line moves across the staves so you always know exactly where you are in the bar.
- Live Note Drawing: As soon as you press keys on your MIDI keyboard, the notes appear right on the staff and grow longer for as long as you hold the keys down.
- Clean Note Lengths: Held notes stay neatly tied to the moving red recording cursor while you play, and snap into finished sheet music the moment you hit Stop.
- Smooth Page Scrolling: The score scrolls smoothly along with your playing and updates the bar and beat counter at the top of the screen.

## 2.4 Right-Click Record Settings: Arming Tracks, Auto-Divisi, Step Input & Count-In

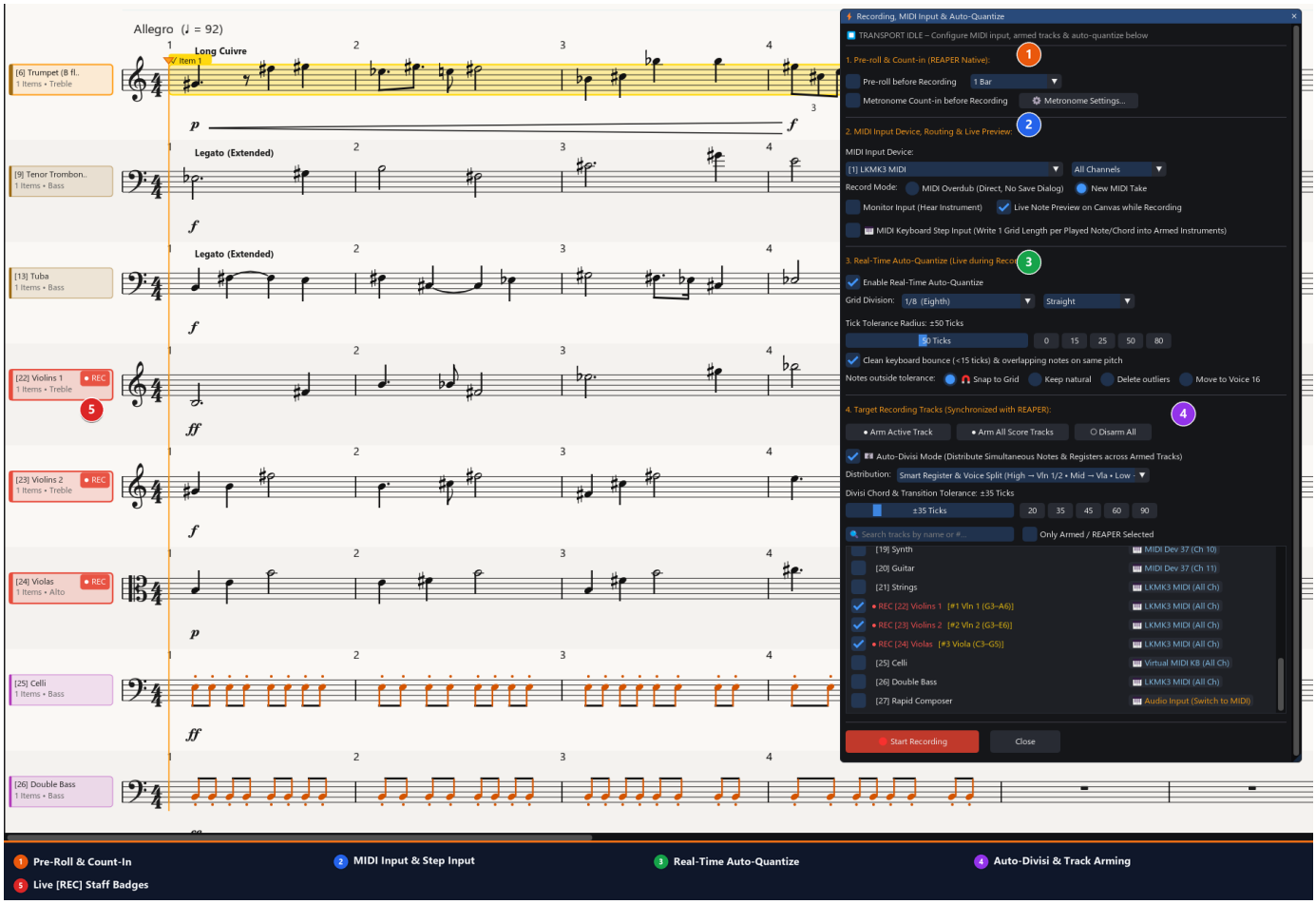



*Figure 2.2: Right-Click Record & Auto-Quantize Modal - Pre-Roll & Count-In (1), MIDI Input & Step Input (2), Real-Time Auto-Quantize (3), Auto-Divisi & Track Arming (4), and Live [REC] Staff Badges (5).*

Right-click the Record button in the top toolbar to open the Recording Settings window:

- One-Click Track Arming: Check the box next to any instrument track in the list to arm it for recording. REAPER-Notator automatically turns on sound monitoring and sets the track to MIDI Overdub mode so you can record right into your score without any annoying audio file pop-ups.
- Red 'REC' Badge on the Staff: Any track you arm for recording gets a clear red border and a red 'REC' badge on the left side of its staff so you never record onto the wrong instrument by mistake.
- Auto-Divisi Live Recording (Play Full Chords Across Multiple Instruments): If you arm two or more instrument tracks and check 'Auto-Divisi Mode', you can play full two-handed chords on your MIDI keyboard and REAPER-Notator will automatically split the notes across your orchestral instruments live as you play:
- Play a Single Note: Only the instrument whose natural range matches that pitch will play and record (for example, playing a low C triggers only the Double Bass).
- Play a Chord (2 or More Notes at Once): The chord is automatically divided from low to high across your armed instruments (for example, low notes go to Celli/Basses, middle notes to Violas, and high notes to Violins).
- Per-Track Mode Button ('A / S / T / B / X'): Next to each track you can click a single letter to choose how that track behaves: 'A' (Auto), 'S' (Smart Range Match), 'T' (Top-Down from highest note), 'B' (Bottom-Up from lowest note), or 'X' (Exclude this track from splitting so it receives every note you play).
- Smooth Legato & Chord Recognition: Even if your fingers don't land at the exact same millisecond when playing a chord, or slightly overlap when playing a melody, the smart transition guard keeps chords together and prevents melodies from jumping to the wrong instrument.
- MIDI Keyboard Step Input (Enter Notes One by One Without a Metronome): Check 'Enable MIDI Keyboard Step Input' when playback is stopped:
- Simply press a note or chord on your MIDI keyboard. REAPER-Notator writes it into the score using your chosen note length (such as a quarter note or eighth note), plays the sound, and automatically moves the cursor forward to the next step.
- Use the 'Step Back' and 'Step Rest' buttons in the window to move backward or insert rests easily.
- MIDI Keyboard & Channel Selection: Choose which plugged-in MIDI keyboard (or REAPER's Virtual MIDI Keyboard) and MIDI

channel you want to use.
- Pre-Roll & Metronome Count-In: Turn on a 1-, 2-, or 4-bar lead-in or metronome clicks before recording begins.

## 2.5 Automatic Timing Cleanup When Recording (Auto-Quantize)

When you play live on a MIDI keyboard, nobody hits every beat with robotic perfection. The Auto-Quantize section inside the Record Settings window cleans up your rhythm automatically so your sheet music looks tidy right away:
- Choose Your Rhythm Grid: Pick the fastest note value you plan to play (Quarter, 8th, 16th, or 32nd notes, plus an optional Triplets checkbox).
- Removes Accidental Double-Hits: Automatically cleans up tiny mechanical key bounces and overlapping notes of the same pitch.
- Timing Tolerance Slider: Adjusts how strict or forgiving the rhythm detector is.
- Four Ways to Handle Off-Beat Notes ('Notes outside tolerance'):

1. '[Snap] Snap to Grid' (Default & Recommended): Gently pulls every recorded note onto the nearest musical beat so your sheet music is immediately clean and easy to read.
2. 'Keep natural (unquantized)': Leaves notes played far off the beat at their original timing.
3. 'Delete outliers': Removes accidental brushed keys that fell far off the beat.
4. 'Move to Voice 16': Moves off-beat notes to background Voice 16 so you still hear them during playback without cluttering Voice 1 on the page.

- Easy Undo: If you don't like a take, simply press Ctrl+Z to undo the entire recording in one step.

## 2.6 Bar, Beat, Time Signature & Tempo Readouts

In the top toolbar next to the transport buttons, you will find three clear displays:
- Bar & Beat Counter (e.g. '004.01.00'): Shows the exact measure and beat where the playhead currently sits.
- Time Signature Box (e.g. 4/4, 3/4, 6/8): Shows the active meter at the playhead. Click it to quickly change the time signature.
- Tempo (BPM) Display: Shows the current speed of the music in beats per minute, updating live during speed-ups (accelerando) and slow-downs (ritardando).

## 2.7 Hearing Notes When You Click Them (Audition Preview)

Whenever the speaker button in the top toolbar is active, REAPER-Notator lets you hear your music as you edit:
- Hear Any Note You Click or Drag: Click a note or drag it up and down the staff to hear its pitch immediately through that track's virtual instrument.
- No False Recording Badges: Clicking a note to hear it never leaves the track stuck in recording mode or shows a red 'REC' badge.
- Page Stays Still While You Edit: Clicking a note, clef, or track name selects that track in REAPER without making your score jump around on screen.
- Plays With the Right Expression: When you click a note, REAPER-Notator checks what articulation (such as Pizzicato or Staccato) and dynamic level are active in that bar so you hear the right sound.

## 2.8 Follow Playhead (Auto-Scroll)

Turn on the Auto-Scroll button in the top toolbar if you want the sheet music to scroll smoothly from left to right during playback, keeping the moving playhead in view at all times.

## 2.9 Automatic Clef Detection (Treble, Bass, Grand Staff, Percussion & TAB)

REAPER-Notator automatically picks the right clef for each instrument track based on its name and note range:
- Standard Orchestral & Band Instruments: Automatically assigns Treble, Bass, Alto, Tenor, or Piano/Harp Grand Staff.
- Smart Drum & Percussion Recognition: Tracks named after unpitched drums (Snare, Kick, Bass Drum, Drum Kit, Taiko, Toms, Hi-Hat, Cymbals, Gong, Tambourine, Shaker, Conga, Bongo, Triangle, etc.) automatically receive a Percussion Clef. Pitched percussion instruments automatically get their proper pitched clefs (Timpani -> Bass Clef; Xylophone, Marimba, Vibraphone, Glockenspiel, Tubular Bells -> Treble Clef; Celesta -> Grand Staff).
- Easy Manual Change Anytime: Want a different clef or Guitar/Bass TAB? Simply right-click the clef at the start of any staff (or open the Clefs panel on the right) and pick the clef you want.

## 2.10 Score Visibility Options: Showing & Hiding Score Layers, Keyswitches & Drawings

Sometimes a full orchestral score has so many markings-dynamics, hairpins, lyrics, chord symbols, articulation badges, and pedal lines-that the page feels crowded. REAPER-Notator lets you hide or show any type of marking with a single click.

Important for Beginners - Hiding Never Deletes Your Music:
Unchecking an item in the Visibility menu only hides it visually on screen so your page looks cleaner. All of your notes, dynamics, pedal movements, and instrument articulations continue to play back 100% normally! Your visibility choices are saved automatically.

1. The '[View] View' Button ('Elements & Layers' Menu in the Top Toolbar - Item 3 in Figure 2.1):
Click the '[View] View' button in the top toolbar to open a checklist of 16 score layers that you can turn on or off individually:

- Sticky Track Names: Keeps each instrument's name label pinned to the left edge of the window as you scroll forward through long pieces, so you always know which line belongs to which instrument.
- Track Name Badges: Shows or hides the colored instrument name boxes on the left side of the staves.
- MIDI Item Boxes: Shows or hides the colored background boxes that indicate where each REAPER MIDI clip starts and ends. (You can also make these boxes lighter or darker in Settings -> Appearance & Colors -> 'MIDI Track Boxes Color Intensity'.)
- Measure Numbers: Shows or hides the bar numbers along the top of the score.
- Dynamic Markers: Shows or hides volume markings like pp, mf, ff, and sfz below the staff.
- Articulations (PC): Shows or hides playing technique labels (such as Legato, Pizzicato, Staccato, or Muted) above the staff.
- Hairpins (< & >): Shows or hides crescendo (<) and diminuendo (>) wedges.
- Text Dynamics (cresc. / dim.): Shows or hides italic text swells like 'cresc. - - -' and 'dim. - - -'.
- Holding / Sustain (Pedal CC64): Shows or hides piano sustain pedal lines (Ped. ... *) below the staff.
- Tempo Markers: Shows or hides speed markings (like Quarter = 120 or Allegro) and gradual speed change lines (accel. / rit.) at the top of the score.
- Octave Lines (8va): Shows or hides octave shift brackets (8va, 8vb, 15ma, 15mb).
- Tuplets / Triplets (3, 5, ...): Shows or hides triplet and tuplet numbers and brackets.
- Text Items ([Text]): Shows or hides free text notes you placed on the page.
- Lyrics Lane ([Lyrics]): Shows or hides sung lyrics beneath vocal staves.
- Chord / Scale Lane: Shows or hides the Chord & Scale bar along the very top of the score.
- Rehearsal Marks Lane: Shows or hides section letters ([A], [B]), bar rehearsal numbers ([1], [2]), and navigation signs (D.C., Segno, Coda).

2. Hiding Unwanted Keyswitch Notes ('Hide Keyswitch Notes'):
Many sample libraries (like Kontakt orchestras or Ample Guitar) use very low or very high piano keys as 'keyswitches' to change sounds. Normally, those trigger notes look like ugly clusters of ledger-line notes on your sheet music. Open the 'Clefs' panel on the right (under 'Filter:') or go to Settings -> General Settings -> 'Hide Keyswitch Notes' to hide them from the page while keeping them working in playback:

- Off (Show All Notes): Shows every note across the entire keyboard.
- Hide all Notes <= C2: Hides low keyswitch notes from C2 downward.
- Hide all Notes <= C1: Hides ultra-low keyswitch notes from C1 downward.
- Hide all Notes >= C6: Hides high trigger notes from C6 upward.
- Ample Guitar Preset (Hide <= C2 & >= C6) / Hide <= C1 & >= C6: Hides both low keyswitches and high strumming trigger keys at the same time.

3. Showing or Hiding Freehand Drawings & Glissando Text:

- Freehand Drawings ('[Layers] Layers' Button): Open the 'Layers' panel on the right to turn all freehand pen markings on or off at once ('Show Annotations on Score'), or click the eye icon next to individual drawing layers per track.
- Glissando & Portamento Labels: Right-click any glissando or portamento line between two notes to show or hide the little italic 'gliss.' or 'port.' word next to the line.

## 2.11 Choosing Which Instrument Tracks Are Visible ('Tracks (X/Y)' Menu)

In the top toolbar, click the '[Score] Tracks (X/Y)' button (Item 5 in Figure 2.1) to choose which tracks from your REAPER project are shown as staves in your score:

- Auto-Detect Tracks (Add New Automatically): Turned on by default so any new MIDI track you create in REAPER automatically appears in your score.
- [Target] Follow REAPER Track Selection (Auto-Switch): When you turn this on, the score automatically switches to show whichever track(s) or MIDI clip(s) you click in REAPER's main window. This is a huge time-saver if you work with large orchestral templates (100+ tracks) and only want to see the instruments you have selected right now.
- Quick Selection Buttons:
- '[OK] Check All': Shows all MIDI tracks in your project at once.
- '[X] Uncheck All': Clears the screen so all tracks are hidden-then simply click any track in the list (or in REAPER) to view just that instrument.
- '[Target] REAPER Sel': Shows only the tracks currently selected in REAPER.
- [Search] Search Box ('Search tracks by name or #...'): Type part of an instrument's name (like 'Horn' or 'Violin') or its track number to find it immediately in large projects.

## 2.12 Working With Multiple Voices & Ghost Notes ('All Notes / Voice 1..16' Menu)

When two melodies share the same staff (for example, Flute 1 and Flute 2 on one staff, or upper and lower hands in a guitar/piano part), musical notation uses separate 'Voices' so the upper melody has stems pointing up and the lower melody has stems pointing down.

Click the '[Voice] All Notes' button in the top toolbar (Item 4 in Figure 2.1) to control voice visibility:

- Switch Between 'All Notes' and 'Voice 1..16': Choose 'All Notes' to work on every voice at once, or pick a specific voice (such as 'Voice 1' or 'Voice 2'). The menu shows how many notes are currently in each voice. When you pick a specific voice, new notes you draw are placed into that voice, and all other voices softly fade into the background ('Ghost Notes') so you can focus.
- Color by Voice: Gives each voice its own distinct color so you can easily tell which note belongs to Voice 1, Voice 2, Voice 3, etc., or uncheck it for classic black-and-white sheet music.
- Hide Inactive Voices Completely: If you prefer not to see faded background voices at all while editing a single voice, check 'Hide Inactive Voices' (in the Voice menu or in Settings -> Voice & Ghost Notes Settings).
- Ghost Notes Fade Slider (5% to 100%): In Settings -> Voice & Ghost Notes Settings, adjust the 'Ghost Notes Opacity' slider to control how faint or visible background voices appear.
- Automatic Voice Splitting ('Auto-Voice'): Click 'Auto-Voice (Whole Track)' or 'Auto-Voice on Selection' right inside the menu to automatically split chords and overlapping melodies into clean upper and lower voices with opposite stem directions.

## 2.13 Settings Window & Dark Mode

Click the gear button in the top toolbar to open the Settings window:

- Neat Category Sections: Everything is organized into clear collapsible sections (General Settings, Portamento & Glissando Defaults, File Paths, Voice & Ghost Notes, Appearance & Colors, MIDI Editor Colors, Mouse Navigation, Keyboard Shortcuts, and Donate).
- Easy Scrolling & Reset Buttons: Scroll smoothly through all options and reset any section to its factory defaults at the bottom of the window.
- Instant Shortcut Search: Type into the search box under 'Customize Keyboard Shortcuts' to find and change any keyboard shortcut in seconds.
- Classic Paper or Dark Mode: Switch between classic warm score paper and eye-friendly Dark Mode, where the page turns dark and all notes, beams, staves, and text flip to crisp light colors.

# Chapter 3: Writing & Editing Notes on the Staff

## 3.1 Clicking to Add Notes & Ledger Lines

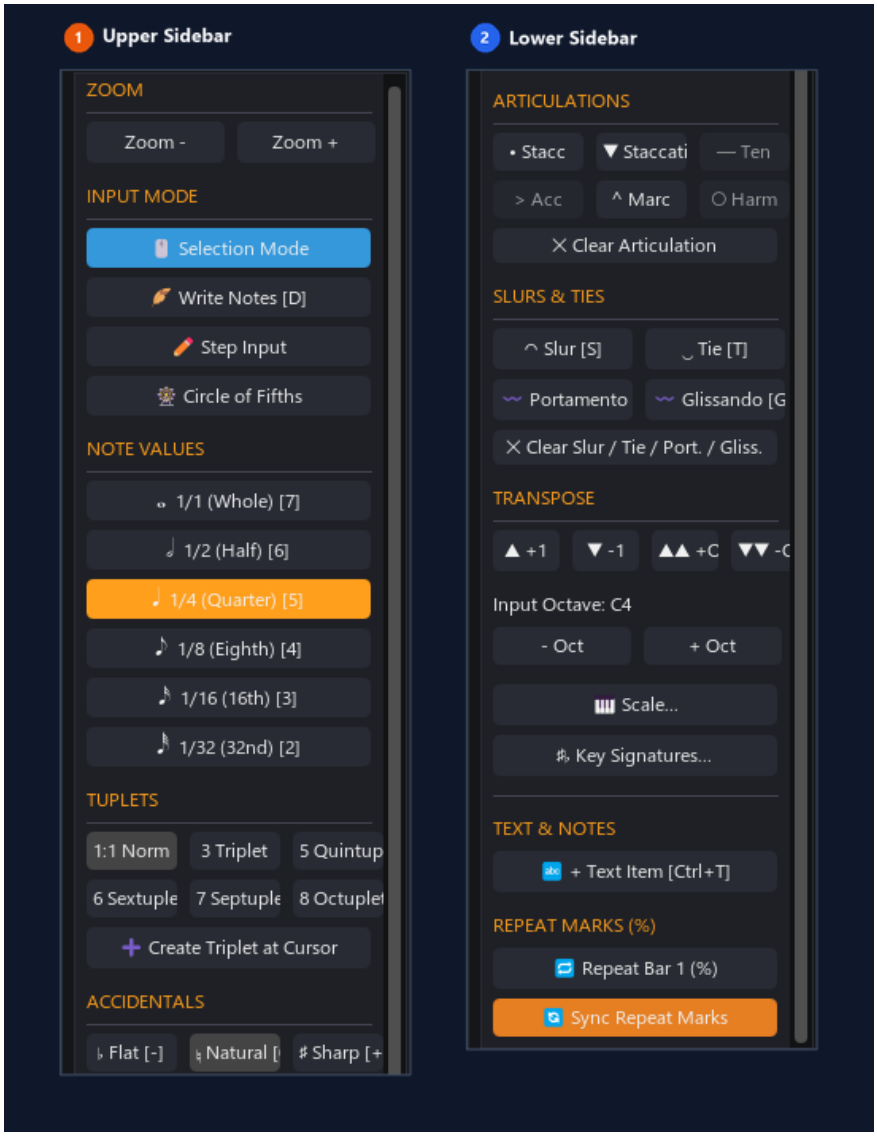



*Figure 3.1: Left Sidebar Palette - Upper Sidebar (Zoom, Input Modes, Note Values, Tuplets, Accidentals) and Lower Sidebar (Articulations, Slurs/Ties/Glissandi, Transpose, Text Items, Repeat Marks).*

How to Add Notes:

- Click Anywhere on the Staff: Select the Pencil (Draw) tool in the left sidebar, pick a note length (like Quarter or Eighth), and click on any line or space of the staff to place a note.
- Automatic Ledger Lines: When you click above or below the 5 lines of the staff, REAPER-Notator automatically draws the extra short lines (ledger lines) for high and low notes.

**DAW Effect:** Immediately creates a real MIDI note in your REAPER track clip so it plays back right away.

## 3.2 Choosing Note Lengths, Converting Selected Notes ('C' Key), Dotted Notes & Grid Snapping

How to Pick Your Rhythm:

- Note Lengths for New Notes (Keys 1 to 6): Click the note buttons in the left sidebar-or press number keys 1 through 6 on your keyboard-to choose Whole (1), Half (2), Quarter (3), Eighth (4), 16th (5), or 32nd (6) for the next notes you draw. Choosing a note value does not accidentally overwrite already-selected notes.
- Convert Selected Notes ('Convert Notes [C]' Button or 'C' Key): Want to change the note length of existing notes on the staff? Select the notes, pick the desired note length (1 to 6), and press 'C' on your keyboard (or click 'Convert Notes [C]' in the left sidebar) to convert all selected notes to the active duration.
- Dotted Notes (Period '.' Key): Click the Dot button in the sidebar (or press the '.' period key) to make a note 50% longer (for

example, turning a 1-beat quarter note into a 1.5-beat dotted quarter note).
- Link Grid to Note Length ('[Link]' Button or Ctrl+K): Click the chain-link button so that whenever you pick a new note length, your rhythmic snapping grid updates automatically to match it.
- Quick Grid Shortcuts (Ctrl+1 to Ctrl+7): Press Ctrl+7 for Whole, Ctrl+6 for Half, Ctrl+5 for Quarter, Ctrl+4 for Eighth, Ctrl+3 for 16th, Ctrl+2 for 32nd, or Ctrl+1 for Free Grid (unquantized placement).

## 3.3 Sharps (#), Flats (b) & Naturals

How to Change Accidentals:
- Flat (b / '-' Key): Lowers the selected note by one half-step.
- Natural ('0' Key): Cancels out a previous sharp or flat in the bar.
- Sharp (# / '=' Key): Raises the selected note by one half-step.

## 3.4 Moving, Stretching, Multi-Note Block Dragging Across Staves & Resizing Clips

How to Adjust Existing Notes:
- Drag Left or Right: Moves a note earlier or later in the bar, snapping cleanly to your chosen beat grid.
- Drag Up or Down: Changes the note's pitch up or down the staff while letting you hear each step.
- Multi-Note Block Drag & Drop Across Staves (With Ghost Notes Preview): Select multiple notes (with Shift+Click or a selection box) and drag any selected note to move the entire phrase together-even directly onto another instrument's staff! While dragging, a glowing Ghost Notes preview shows the exact target notes, stems, ledger lines, and accidentals. Hold Ctrl while dragging (Ctrl + Drag) to copy the phrase instead of moving it.
- Move Selected Notes to Staff Above or Below (Ctrl + Up / Down Arrows): Press Ctrl+Up or Ctrl+Down to move all selected notes at their exact beat positions to the next visible instrument staff above or below.
- Drag the Right Edge of a Note: Pulls the note longer or shorter.
- Keyboard Arrow Keys: Press Up/Down Arrows to nudge pitch by 1 half-step (or Shift+Up/Down for a full octave). Press Left/Right Arrows to move notes earlier or later by one grid step (or Shift+Left/Right to make them shorter or longer).
- Resize MIDI Clips Directly on the Score: Drag the left or right border of any colored MIDI clip box right on the sheet music to make the clip longer or shorter in REAPER.

## 3.5 Selecting Multiple Notes & Deleting Cleanly

How to Work With Groups of Notes:
- Draw a Selection Box: With the Select (arrow) tool active, click and drag an empty area of the page to draw a rectangle around multiple notes-even across several bars or staves.
- Shift + Click: Hold Shift and click individual notes to add or remove them from your selection.
- Delete / Backspace Key: Deletes all selected notes at once and automatically cleans up any attached articulations or ties so nothing is left behind.

## 3.6 Notes Next to Each Other in Chords & Voice Stems

Automatic Sheet Music Tidying:
- Notes One Step Apart (Seconds): When two notes in a chord sit right next to each other on neighboring line-and-space positions, REAPER-Notator automatically places one notehead slightly to the side of the stem so they never overlap into a black blob.
- Automatic Stem Directions for Voices: When you use multiple voices on one staff, Voice 1 stems automatically point up and Voice 2 stems point down.

## 3.7 Automatic Rests

You never have to draw rests by hand if you don't want to:
- Empty Bars: Any bar without notes automatically shows a centered whole-bar rest.
- Musical Rests Between Notes: Gaps between notes are automatically filled with the proper quarter, eighth, or 16th rests that match the beats of your time signature.
- Clean Multi-Voice Rests: When two voices share the same rest at the same time, they merge into a single clean rest so the staff stays easy to read.

## 3.8 Stepping Through Notes With Your Keyboard (No Mouse Needed)

How to Navigate Notes by Keyboard:
- Alt + Left / Right Arrows: Jumps from note to note backward or forward through your score without touching the mouse.
- Alt + Shift + Left / Right Arrows: Expands your selection to include the next or previous notes.
- Follows Your Selection: When Auto-Scroll is turned on, the page automatically glides along as you step from note to note.

## 3.9 Drum & Percussion Staves and Custom Drum Noteheads

Writing for Drums & Percussion Made Easy:

- Clean Percussion Staves: On any staff using a Percussion Clef, sharps, flats, and key signatures are automatically hidden so your drum part looks like a real drum chart.
- Standard Drum Kit Layout: Clicking or dragging notes on a Percussion Clef staff automatically plays and places standard drum kit sounds on their familiar staff lines and spaces:
- Bottom Space (F4): Kick / Bass Drum.
- 2nd Line (G4): Low Floor Tom.
- 2nd Space (A4): High Floor Tom / Pedal Hi-Hat.
- Middle Line (B4): Low Tom.
- 3rd Space (C5): Snare Drum / Side Stick / Hand Clap.
- 4th Line (D5): Mid Tom.
- 4th Space (E5): High Tom / Cowbell.
- Top Line (F5): Ride Cymbal / Closed & Open Hi-Hat / Tambourine.
- Space Above Staff (G5): Ride Cymbal 2 / China Cymbal.
- 1st Ledger Line Above Staff (A5): Crash Cymbal / Splash Cymbal.
- Works Across All Octaves: Even if your drum library puts drums in very low or very high octaves, REAPER-Notator keeps them neatly inside the drum staff without endless ledger lines.
- Changing Drum Notehead Shapes (Cross, Diamond, Triangle, Slash, Ghost Note):
- By default, new notes use standard round/oval noteheads.
- Right-click any note (or group of selected notes) and open '[Drum] Notehead / Percussion Head' to pick from 9 classic percussion notehead shapes:

1. Standard (Default Oval)
2. Cross (X) - Great for Hi-Hat and Cymbals
3. Circled Cross - Open Hi-Hat or Crash Cymbal
4. Diamond - Ride Bell, Rim, or String Harmonics
5. Triangle Up - Triangle, Woodblock, or Cowbell
6. Slash - Rhythm Section Chords or Rimshots
7. Circle / Open - Tambourine or Open Drum Tone
8. Plus (+) - Muted or Stopped Note
9. Ghost / Parenthesized - Soft Ghost Note in parentheses `( )`

## 3.10 Guitar & Bass Tablature (TAB) & Hiding Guitar Keyswitches

Writing for Guitar & Bass:

- Six Ready-to-Use TAB Layouts: Open the 'Clefs' panel on the right (under 'Tablature & Fretted') to switch any guitar or bass track to:
- '6-String Guitar TAB' (Standard tuning E-A-D-G-B-E) or 'Notation + 6-String TAB' (both standard sheet music and TAB linked together).
- '4-String Bass TAB' (E-A-D-G) or 'Notation + 4-String TAB'.
- '5-String Bass TAB' (Low B-E-A-D-G) or 'Notation + 5-String TAB'.
- Automatic & Manual String/Fret Numbers: REAPER-Notator automatically calculates comfortable fret and string numbers for every note you play or draw. Prefer to play a note on a different string? Simply right-click the note and choose 'String / Fret Assignment' to pick the exact string you want.
- Hide Virtual Guitar Keyswitches: Use the 'Keyswitch Filter' right inside the Clefs panel (such as the 'Ample Guitar Preset') to hide low keyswitch notes and high strumming trigger keys from the TAB while keeping them active in playback.

# Chapter 4: Automatic Note Beaming & Stem Direction

## 4.1 Automatic Beat Grouping by Time Signature

When you write eighth, 16th, or 32nd notes next to each other, REAPER-Notator automatically connects their stems with solid beams grouped to match your time signature (for example, grouping 4/4 bars into clear beats and 6/8 bars into two groups of three eighth notes).

## 4.2 Natural Beam Angles

Beams gently tilt upward when a melody rises and downward when a melody falls, while staying level and tidy when notes jump back and forth.

## 4.3 Dotted Rhythms & Short Beam Stubs

Mixed rhythms-like a dotted eighth note followed by a single 16th note-automatically display a neat short beam stub pointing inside the beat group so the rhythm is effortless to sight-read.

## 4.4 Flipping Stems Up or Down ('X' Key)

Select any note or beamed group and press the 'X' key on your keyboard (or click the stem flip button in the sidebar) to flip its stem direction between pointing up and pointing down.

## 4.5 Crisp, Smooth Beams at Every Zoom Level

All note stems, beams, and tuplets are drawn as sharp vector graphics that stay smooth and clean whether you zoom out for a full orchestral overview or zoom in to 200%.

# Chapter 5: Dynamics, Hairpins (Crescendo/Diminuendo) & Bow Swells

## 5.1 One-Click Volume Markings (ppp to fff & Accents)

Open the 'Dynamics' panel on the right side of the window to place standard musical volume markings below any staff with a single click:

- ppp (Very, very soft) -> Softest playback (~20-25)
- pp (Very soft) -> (~35-40)
- p (Soft) -> (~50-55)
- mp (Moderately soft) -> (~65-68)
- mf (Moderately loud) -> (~80-82)
- f (Loud) -> (~95-98)
- ff (Very loud) -> (~110-112)
- fff (Very, very loud) -> Maximum power (~125-127)
- sfz, rfz, fz, fp, sfp (Sudden Dynamic Accents): Creates a strong initial attack on the note followed immediately by the right sustain or drop to soft piano-working automatically for both virtual instruments and Synthesizer V vocals.

**DAW Effect:** Places the dynamic symbol under your staff and automatically adjusts your note velocities and expression curves in REAPER.

## 5.2 Choosing Which MIDI Controller Shapes Your Volume

Different sample libraries use different controllers for volume and expression. In the Dynamics panel, you can choose what your dynamics control:

- CC1 (Modulation Wheel): The standard volume/timbre controller for most orchestral string, brass, and woodwind libraries.
- CC11 (Expression): Secondary volume controller used by many libraries.
- CC7 (Main Volume): Overall channel volume.
- Velocity Only: Adjusts how hard notes are struck (ideal for piano, harp, drums, and plucked instruments).

**DAW Effect:** Draws smooth controller curves directly into your REAPER MIDI clip.

## 5.3 Crescendo (<) & Diminuendo (>) Hairpin Wedges

How to Create Smooth Volume Swells:

- Crescendo (<): Gradually gets louder from the start of the wedge to the end.
- Diminuendo (>): Gradually gets softer from the start of the wedge to the end.
- Easy Drag Handles: Click any hairpin on the page to reveal two round handles at its start and end-drag them left or right to stretch or shorten the swell.
- Curve Shape Choice: Choose between a steady straight climb (Linear) or a natural dramatic swell (Exponential).
- Never Blocked by Staccato Dots: You can freely stretch hairpins across bars full of staccato dots, accents, or tenutos without the hairpin getting stuck or squeezed.

## 5.4 Automatic Bow Swells on Long Notes

Real string, brass, and woodwind players rarely hold a long note completely flat-they gently swell into the note and taper off:

- Bow Swell Toggle: Automatically creates a natural rise-and-fall expression curve over sustained notes.
- Peak Position Slider (0.0 to 1.0): Choose where the high point of the swell happens. Leave it at 0.5 for a balanced center swell, move it to 0.8 for a dramatic late build, or set it to 0.2 for an early push.

## 5.5 Temporarily Turning Off Dynamic Shaping (Bypass)

Want to keep your dynamic markings visible on the page for printing or reading, but stop them from changing your MIDI controller curves? Simply check the Bypass box in the Dynamics panel.

# Chapter 6: Playing Techniques (Articulations), Slurs, Ties & Reaticulate

## 6.1 Automatic Sound Bank Discovery & Choosing an Instrument Bank

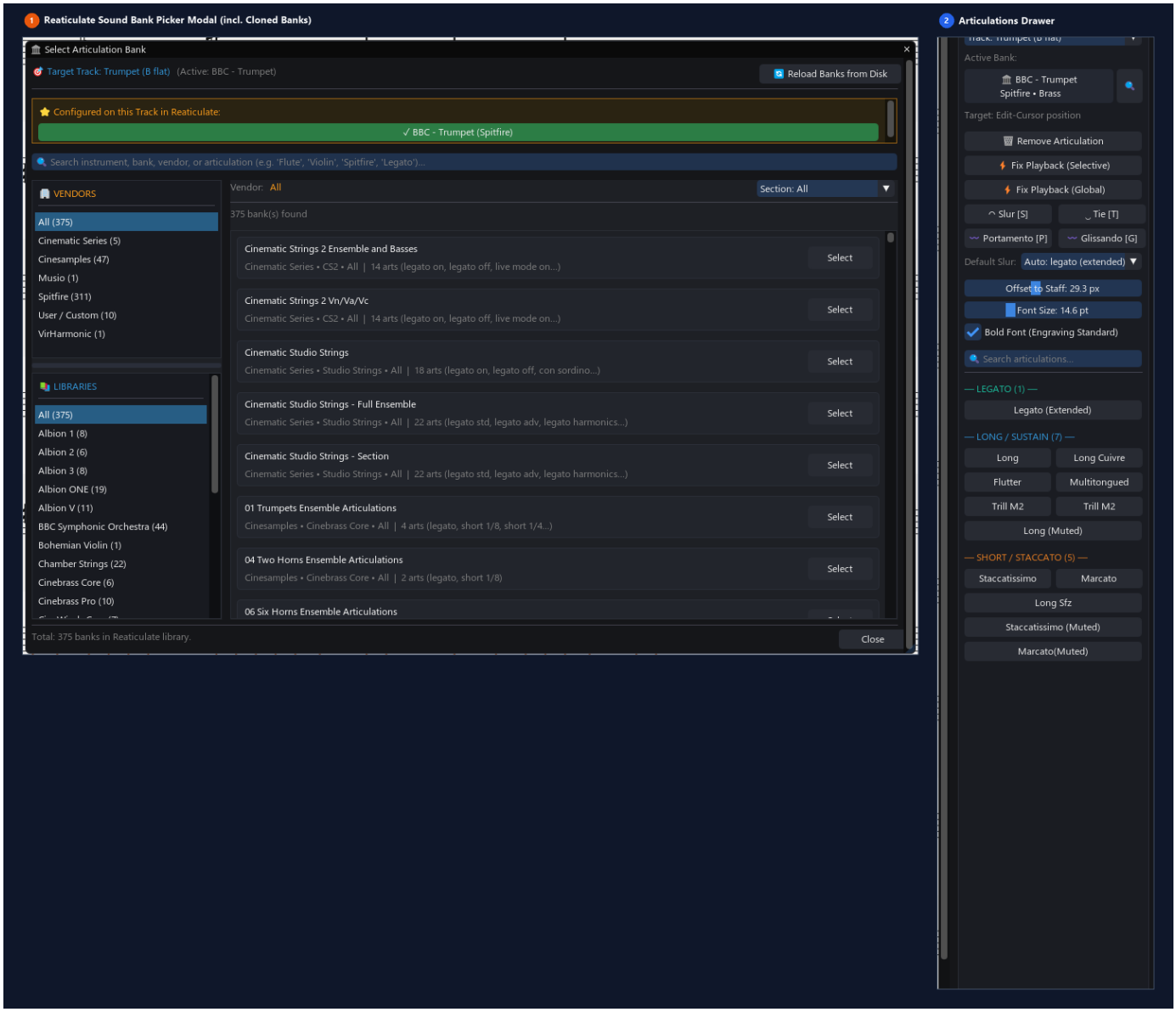



*Figure 6.1: Articulations Drawer & Reaticulate Sound Bank Picker - Target Track (1), Active Sound Bank (2), Search & Configured Banks (3), Vendor/Library Filter (4), and 1-Click Bank Assignment (5).*

How REAPER-Notator Connects to Your Sample Libraries:

- Automatic Sound Bank Matching: When you click on an instrument track, REAPER-Notator automatically looks at your installed Reaticulate sound banks and matches the track's name (like 'Violin I', 'Cello', 'Horns', or 'Trumpet') to the right articulation list.
- Choosing or Changing a Sound Bank Anytime: Open the 'Articulations' panel on the right and click the magnifying-glass / bank button to open the Sound Bank Picker (shown in Figure 6.1). Search for your library (such as Spitfire, BBC Symphony Orchestra, Orchestral Tools, Cinematic Studio, or VSL) and click any bank to assign it to your track in one click.

## 6.2 Zero-Setup New Tracks: Automatic Reaticulate Plugin Placement

You never have to worry about setting up Reaticulate manually on new tracks:

- Automatic Plugin Insertion: If you create a brand-new track in REAPER with your virtual instrument (like Kontakt, Sine, Opus, or BBC SO) and pick a sound bank-or click any articulation in REAPER-Notator-Notator automatically adds the Reaticulate plugin to that track for you.
- Always Placed in the First FX Slot: For Reaticulate to trigger your instrument's keyswitches properly, it needs to sit in the very first FX slot before your virtual instrument. REAPER-Notator automatically puts it in Slot 1 so your articulations work right away.
- Clean Bank Switching: When you switch a track to a different sound bank, the old bank is cleanly replaced so you never end up

with duplicate or conflicting banks stacked on top of each other.

## 6.3 Adding Playing Techniques (Staccato, Pizzicato, Legato, Accents & More)

How to Add Articulations to Your Notes:

- Quick Buttons in Sidebar & Articulations Drawer: Click any playing technique to place it on your music:
- Short & Accented Marks: Staccato (dot), Staccatissimo (wedge), Accent (>), Marcato (^), Tenuto (-).
- Instrument Techniques: Legato, Long / Sustain, Pizzicato (pizz.), Arco, Muted (con sord.), Sul Ponticello, Sul Tasto, Col Legno, Tremolo, Trills, and Harmonics (o).
- Automatic Above/Below Placement: Dots, accents, and wedges automatically sit on the clear side of the notehead opposite the stem, avoiding staff lines so they are always easy to read.
- Three Ways to Apply an Articulation:

1. On Selected Notes: Select one or more notes on the staff and click an articulation button.
2. At the Edit Cursor: If no notes are selected, place REAPER's Edit Cursor on a note and click the articulation.
3. On an Empty Space: If you click an articulation where no note exists yet, REAPER-Notator places a technique switch badge above the staff at the cursor so the next notes you write or record will use that sound.

## 6.4 How Short Articulations Automatically Return to Normal Sustain

Why You Don't Have to Reset Keyswitches Manually:

- Short Marks vs. Long Techniques: Markings like Staccato, Staccatissimo, and Accent only apply to the specific notes that have the dot or accent mark. Once those notes finish playing, a real player naturally goes back to normal sustained playing-and REAPER-Notator makes your virtual instrument do the exact same thing automatically!
- Automatic Return to Normal Sound: Whenever you put a Staccato or Accent on a note (or a whole group of notes), REAPER-Notator switches your instrument to Staccato for those notes and automatically switches it back to your normal Long / Sustain sound right after the last staccato note ends.
- Smart Cleanup When You Delete Notes: If you delete some staccato notes, or click 'Remove Articulation', REAPER-Notator automatically updates the sound switches so your instrument never gets stuck in Staccato.

## 6.5 Fixing Stuck or Copied Sounds with 'Fix Playback' (Selective vs. Global)

When Should You Click 'Fix Playback'?

- Watch the Video Guide: See 'Fix Playback', Ties, and Slurs in action here:
- 'Reaper-Notator - Infos about Fix Playback , Ties and Slurs function': https://youtu.be/ahpEfmgGb3M
- Why Sounds Can Get Mixed Up When Copying Clips Between Tracks: In REAPER, it is very common to copy a MIDI clip from one track to another (for example, copying a Violin melody down to the Cellos or Flutes). However, your Violin library and your Flute library might use completely different switch numbers for Staccato and Legato-or the Flute might not have a Pizzicato sound at all! As a result, the copied clip might play the wrong technique or stay silent.
- The One-Click Solution ('[Fix] Fix Playback'): Open the 'Articulations' panel on the right and choose between two buttons:
- 'Fix Playback (Selective)': Fixes ONLY the single MIDI clip you currently have selected (or the clip under your cursor), leaving the rest of your project untouched.
- 'Fix Playback (Global)': Scans and refreshes EVERY track and MIDI clip across your entire project in one click-perfect after copying lots of parts around an orchestral template.
- What 'Fix Playback' Does Automatically:

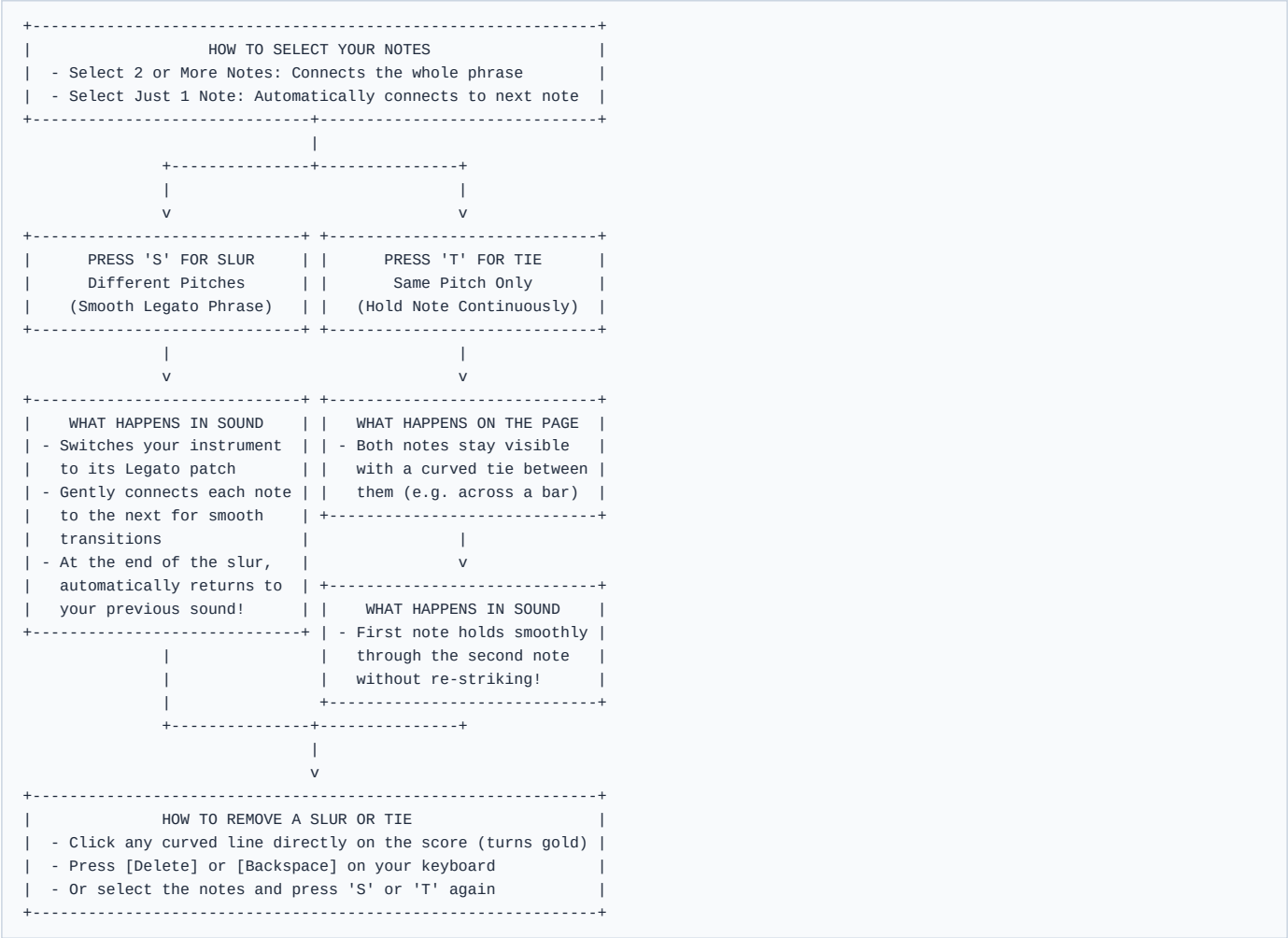
1. Reads the visual articulation marks on your sheet music.
2. Looks up the right sound switches for the current track's instrument library.
3. If the new instrument doesn't have a certain sound (for example, if you copied a string Pizzicato part onto a Flute or Piano), it removes the unsupported switch and sets the track cleanly to its normal playing sound.
4. Cleans up tiny accidental note overlaps and refreshes dynamic hairpins.

- Emergency Reset (Right-Click 'Fix Playback (Selective)'): If a MIDI clip you imported from somewhere else has messy controller data and you want a completely fresh start, right-click 'Fix Playback (Selective)' and click '[ Purge All Articulations (Last Resort) ]' to strip all articulation switches from that clip and reset it to clean normal sustain.

## 6.6 Slurs (Smooth Legato Phrases) & Ties (Held Notes)

How Slurs and Ties Work:

- Video Tutorial: Watch practical examples of Ties and Slurs here:
- 'Reaper-Notator - Infos about Fix Playback , Ties and Slurs function': https://youtu.be/ahpEfmgGb3M




- What Is the Difference Between a Tie and a Slur?
- Tie ('T' Key): Connects two notes of the EXACT SAME pitch (for example, holding a C across a barline into another C). In playback, the note is struck once and held continuously for the combined length of both notes without striking the second note again.
- Slur ('S' Key): Draws a smooth curve over two or more notes of DIFFERENT pitches. It tells your virtual instrument to switch to its connected Legato sound and smoothly glide from pitch to pitch. When the slur ends, your instrument automatically returns to whatever sound it was using before the slur.
- How to Add a Slur or Tie:
- Select 1 note (or multiple notes) and press 'S' for a Slur or 'T' for a Tie-or click the 'Slur [S]' / 'Tie [T]' buttons in the left sidebar.
- Helpful Safeguard: If you accidentally press 'T' (Tie) on two notes that have different pitches, REAPER-Notator will politely remind you to use 'S' (Slur) instead.
- How to Delete a Slur or Tie:
- Simply click the curved line right on the score (it will highlight in gold) and press Delete or Backspace-or select the notes and press 'S' or 'T' again.

## 6.7 Realistic Glissandos, Repeat Bar Signs (%) & Moving MIDI Clips

More Time-Saving Score Features:

- Realistic Glissandos (Harp, Piano, Strings & Trombone Slides): Select a starting note and click 'Glissando' in the left sidebar. On the page, you see a clean wavy or straight glissando line; in playback, REAPER-Notator plays all the intermediate scale or chromatic notes smoothly without cluttering your sheet music with dozens of tiny notes.
- One-Bar Repeat Signs ('%'): Place a '%' Repeat Bar sign in any measure to automatically repeat the previous bar's notes and articulations without copying and pasting. You can even chain several '%' bars in a row, and if you change a note or articulation in the first bar, all the repeated bars update automatically!
- Dragging MIDI Clips in REAPER Moves Everything Together: Whenever you drag a MIDI clip left or right to a new bar in REAPER's main timeline, all of your hairpins, dynamics, pedal lines, lyrics, slurs, ties, and text markings automatically move along with it.

## 6.8 Automatic Articulation Delay Calibration (Keeping Slow Attacks in Time)

Why Do Some Orchestral Samples Sound Late?
In many realistic orchestral libraries, slow lush legato strings or brass swells take 50 to 150 milliseconds to build up sound after a note starts, while fast staccato notes speak almost instantly. Normally, people have to drag legato notes early off the beat so they sound in time-which ruins how the sheet music looks!

How REAPER-Notator Solves This Without Messing Up Your Sheet Music:

- Off by Default (You Are Always in Control): This feature is turned off by default on every track so nothing changes unless you want it to.
- How to Turn It On & Scan Your Instrument:

1. Right-click any instrument's Track Name Badge on the left side of the score.
2. Click 'Scan / Calibrate Track Articulation Delays...'.
3. Click '[ Start Automatic Articulation Scan ]'. REAPER-Notator will briefly play a single test note through each articulation in your track's sound bank, listen to how many milliseconds each sound takes to speak, and save the results.
4. Right-click the Track Name Badge and turn 'Track Articulation Negative Delay' ON.

- Live Timing Pill Next to the Track Name: Once turned on, you will see a small badge next to the instrument name (such as `Legato (-85ms)` or `Staccato (-25ms)`). Your notes stay 100% on the beat on the sheet music page, while REAPER-Notator triggers each articulation just early enough behind the scenes so every note lands right in the pocket! Click the little ms badge anytime to fine-tune the timing with a slider.

# Chapter 7: Piano Sustain Pedal Markings (Ped. ... *)

## 7.1 Adding Sustain Pedal Lines

Open the 'Dynamics / Pedal' panel or use the Pedal tool to place piano sustain pedal markings cleanly beneath the bass staff. REAPER-Notator automatically keeps pedal lines below low notes and hairpins so they never overlap.

## 7.2 Three Pedal Line Styles

Choose the look you prefer:

- Classic Style: Traditional 'Ped.' sign at the start, a dashed line while held, and an asterisk ('*') where the pedal is lifted.
- Bracket Style: Clean modern horizontal line with crisp upward hooks at the start and end.
- Notch / Retake Style: Shows a quick '^' notch wherever the player briefly lifts and re-presses the pedal.

## 7.3 Adjusting Pedal Length & Adding Quick Pedal Lifts (Retakes)

How to Edit Pedal Lines:

- Drag Start & End Handles: Click any pedal line and drag its round start or end handle left or right.
- Right-Click to Add a Pedal Lift (Retake): Right-click anywhere along a pedal line to add a quick lift-and-repress notch where the harmony changes.

**DAW Effect:** Automatically writes MIDI Sustain Pedal (CC64) on/off messages into your track.

# Chapter 8: Tempo Markings, Accelerando/Ritardando & Time Signatures

## 8.1 Adding Tempo Markings (Adagio, Allegro, BPM)

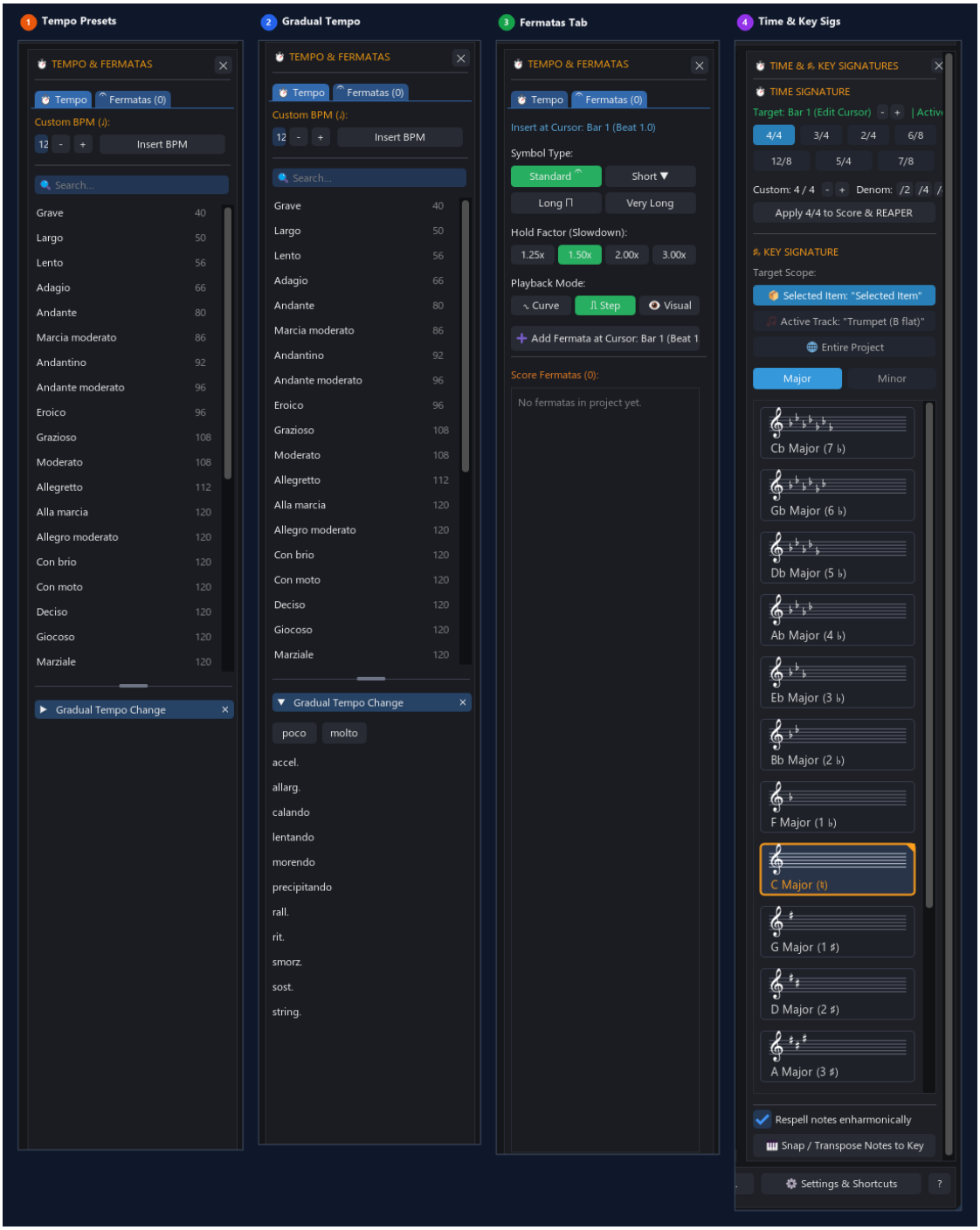



*Figure 8.1: Tempo Presets (1), Gradual Tempo Transitions (2), Score-Wide Fermatas Tab (3), and Time & Key Signatures Drawer (4).*

Open the 'Tempo' panel on the right to place classical tempo markings (like Adagio, Andante, Moderato, Allegro, or Presto) with your exact beats-per-minute speed (e.g. Quarter = 120) at the top of the score.

**DAW Effect:** Immediately updates REAPER's master project tempo at that bar.

## 8.2 Gradual Speed Changes (Accelerando & Ritardando)

Use the Gradual Tempo Transitions section in the Tempo panel to place 'accel.' (speeding up) or 'rit.' / 'rall.' (slowing down) lines across several bars.

**DAW Effect:** Smoothly ramps REAPER's playback tempo up or down across those measures.

## 8.3 Double-Click any Tempo Mark to Change Its Speed

Want to try a slightly faster or slower tempo? Simply double-click any tempo marking right on the top of your score, type the new BPM number, and press Enter.

## 8.4 Changing Time Signatures in the Middle of a Piece

How to Change the Meter at Any Bar:

- Place REAPER's cursor in the bar where you want the new meter to start (for example, Bar 5), open the 'Key & Time Signatures' panel on the right (or click the Time Signature readout in the top bar), and click any time signature (4/4, 3/4, 2/4, 6/8, 12/8, 5/4, 7/8, 9/8, or a custom meter).
- Displayed Across All Staves: The new time signature appears cleanly at the start of that bar on every instrument staff, while all earlier bars keep their original time signature.
- Synced With REAPER's Timeline: REAPER's main bar ruler and metronome automatically update to match your score.

## 8.5 Two-Way Sync With REAPER's Tempo Envelope & Imported MIDI Files

If you import a MIDI file that already has tempo changes-or if you adjust tempo points directly on REAPER's master tempo track-REAPER-Notator automatically shows those tempo changes and gradual speed-ups/slow-downs right on your score.

# Chapter 9: Octave Lines (8va, 8vb, 15ma, 15mb)

## 9.1 Making High and Low Notes Easy to Read

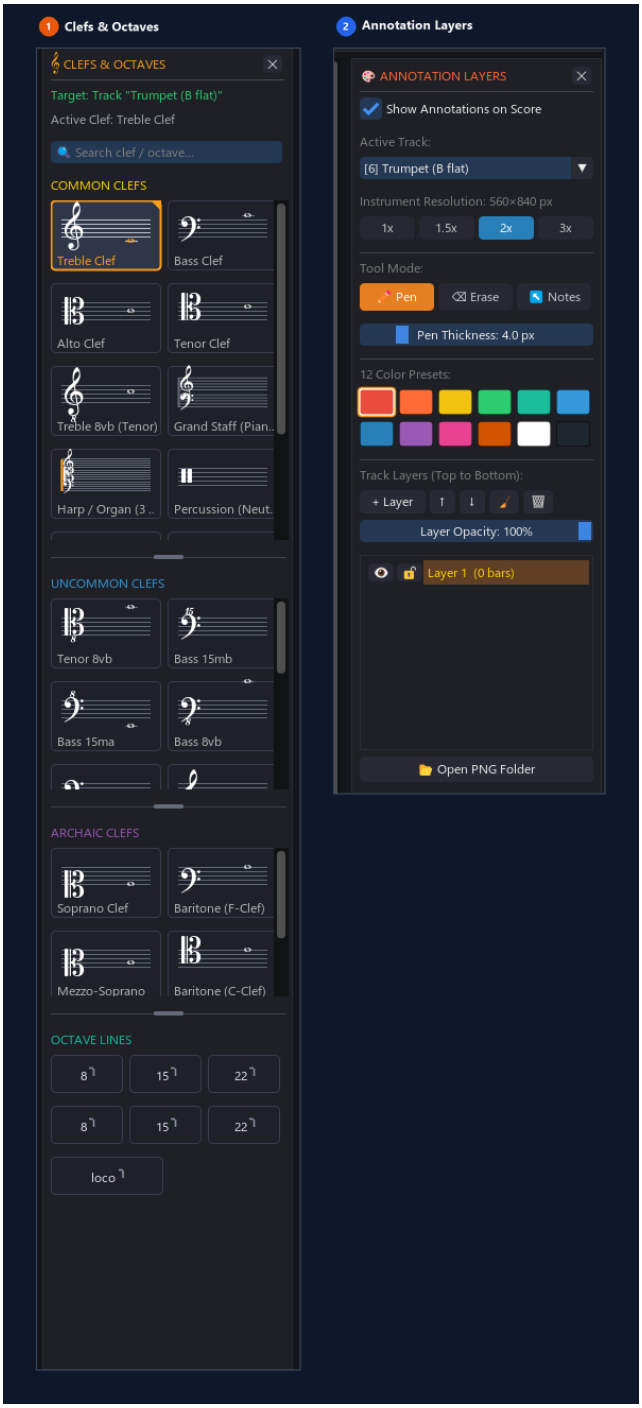



*Figure 9.1: Clefs & Octave Lines Drawer (Common, Uncommon & Archaic Clefs, 8va/15ma/22ma/loco) and Freehand Annotation Layers Drawer.*

When a melody goes very high or very low, counting lots of extra ledger lines above or below the staff gets tiring. Open the 'Clefs' panel on the right to place an Octave Line over those notes so they stay comfortably inside the staff:

- 8va: Notes play 1 octave higher than written.
- 8vb: Notes play 1 octave lower than written.
- 15ma: Notes play 2 octaves higher than written.
- 15mb: Notes play 2 octaves lower than written.

## 9.2 Easy to Read on the Page, True High/Low Pitch in Sound

**DAW Effect:** Your notes sit comfortably on the staff where they are easy to read, while your virtual instrument plays the true high or low octave.

# Chapter 10: Orchestral Pattern Browser (1,500 Ready-to-Use Musical Ideas)

## 10.1 Browse 1,500 Musical Patterns Across 10 Categories

Need instant inspiration or a driving accompaniment? Open the 'Patterns' panel on the right to browse 1,500 ready-made musical phrases organized into 10 categories:

- 1. Strings Staccato (150x driving action ostinatos and rhythmic pulses)
- 2. Strings Pizzicato (150x light plucked grooves and playful patterns)
- 3. Brass Blockbuster (150x bold horn fanfares and low brass power chords)
- 4. Cinematic Melodies (150x expressive main themes)
- 5. Counter Melodies (150x secondary lines that weave around a lead melody)
- 6. Woodwinds Textures (150x agile runs and shimmering figures)
- 7. Cinematic Piano (150x flowing arpeggios and emotional ballads)
- 8. Ancient Greek & Roman Harp (150x modal harp hymns and processions)
- 9. Orchestral & Cinematic Percussion (150x Timpani, Taiko, Bass Drum, Snare, Toms, Anvil, and Mallets)
- 10. Low Strings - Celli & Double Bass Tension (150x suspenseful low string motors and thriller grooves)
- 11. Custom User Patterns (Save your own favorite musical phrases for reuse anytime!)

## 10.2 Listening to Patterns, Dragging Into Your Score & Saving Your Own

How to Use the Pattern Browser:

- Preview Before You Insert: Click the play button on any pattern card to hear the entire multi-bar phrase played live through your chosen instrument track.
- Drag & Drop Right Onto the Staff: Drag any pattern from the browser directly onto your score-a golden preview shows you exactly where the notes will land before you let go of the mouse.
- Save Your Own Phrases: Select any MIDI clip you wrote in REAPER and click 'Capture REAPER Item' to add it to your personal Custom User Patterns collection.

## 10.3 One-Click Pattern Library Download

If the pattern library isn't installed yet, simply click the 'Download Factory Library' button inside the Patterns panel to download and unpack all 1,500 patterns in about one second without leaving REAPER.

# Chapter 11: Rehearsal Marks, Fermatas, Arranging Tools, Melodic Pathfinder, Circle of Fifths & Lyrics

## 11.1 Section Letters ([A], [B]), Custom Markers & Quick 'Ctrl+M' Shortcut

Keep your score organized into clear musical sections (like Intro, Verse, Chorus, or Rehearsal Letters [A], [B], [C]):

- Top Rehearsal Lane: Sits right above the bar numbers so you can see your song structure at a glance.
- Drag & Drop Anywhere: Click and drag any rehearsal mark left or right to move it to another bar.
- Automatic Letter & Number Counting: Standard rehearsal marks count up automatically as [A], [B], [C]... or [1], [2], [3]... If you insert a new letter in the middle of the piece, all later letters update automatically.
- Double-Click to Rename ('Intro', 'Climax', 'Solo'): Double-click any rehearsal mark on the page (or right-click it and choose 'Rename Mark') to type your own custom label.
- Press 'Ctrl+M' Anytime (Even While Playing or Recording!): Press Ctrl+M on your keyboard to drop a numbered marker ('Marker 1', 'Marker 2'...) right at the moving playhead or edit cursor.
- Classic Repeat & Jump Signs: One-click buttons in the Tools panel let you place Da Capo (D.C.), D.C. al Fine, Dal Segno (D.S.), D.S. al Coda, Segno, Coda, and Fine signs.

## 11.2 Fermatas (Hold / Pause Signs Across All Instruments)

Place a classical Fermata (bird's-eye hold sign) when you want the whole ensemble to hold a note or chord:

- Four Fermata Styles: Choose from Standard, Short (triangular), Long (square), or Very Long fermatas in the Tempo / Fermatas panel.
- Clickable on Every Staff: Click any fermata on any instrument staff to select or adjust it.
- Automatic Playback Hold: During playback, REAPER-Notator gently stretches the tempo under the fermata (from 1.25x to 3.0x longer) so your instruments naturally hold the pause and resume normal speed afterward.
- Easy Removal: Click a fermata and press Delete or Backspace to remove it.

## 11.3 The Score Tools Panel (Quick Musical Helpers)

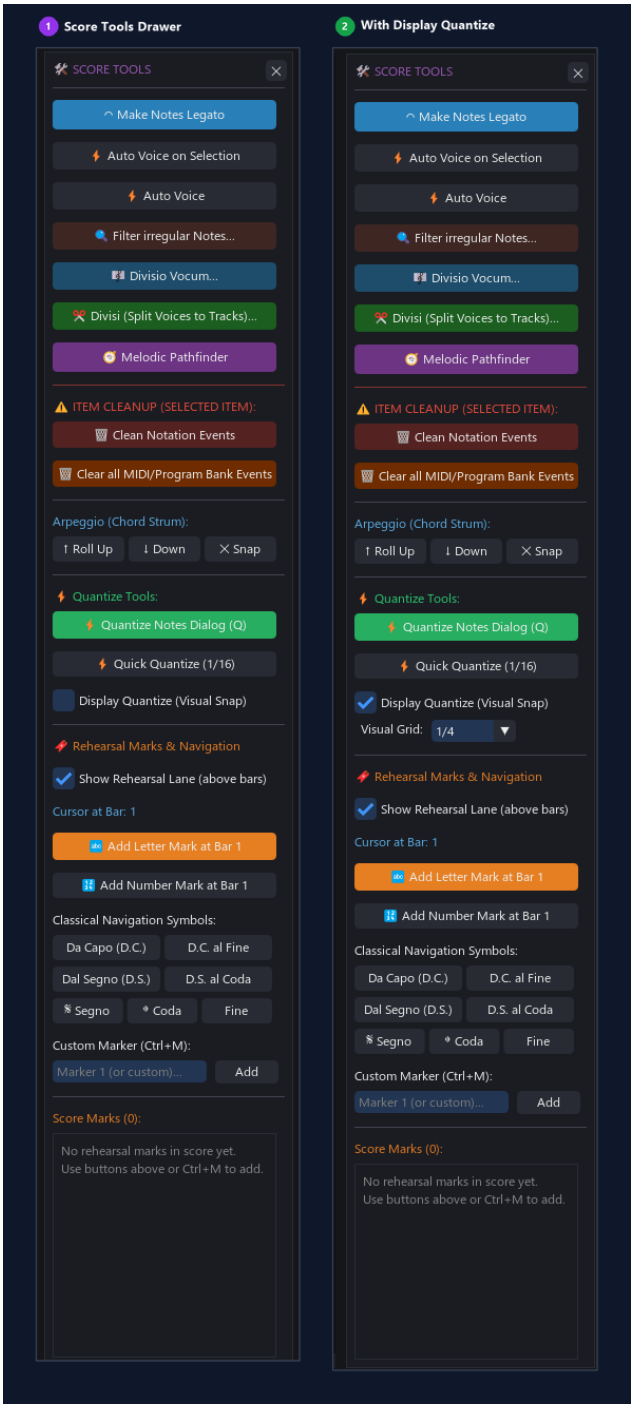



*Figure 11.1: Score Tools & Rehearsal Marks Drawer - Standard View (1) and with Display Quantize / Visual Snap Active (2).*

Open the 'Tools' panel on the right for one-click musical helpers:

- Make Notes Legato: Stretches selected notes so each note lasts right up to the start of the next note without gaps.
- Auto-Voice & Auto-Voice on Selection: Automatically splits chords and overlapping melodies on a single staff into an upper melody (Voice 1, stems up) and lower melody (Voice 2, stems down).
- Filter Irregular Notes: Finds notes from live keyboard playing that landed slightly off the beat, highlights them in bright yellow on the score, and lets you select them or move them to a background voice.
- Divisio Vocum (Spread Chords Across Orchestra Tracks): Takes chords played on one track (such as a piano sketch) and spreads the notes from top to bottom across your orchestral tracks below (Violins, Violas, Cellos, Basses).
- Divisi (Split Voices onto Cloned Instrument Tracks): Takes a staff that has multiple voices (Voice 1, Voice 2, Voice 3) and creates ready-to-play duplicate tracks in REAPER with the exact same virtual instrument and sound settings for each voice.
- Arpeggio Strums (Harp, Guitar & Piano Rolls): Draws a classic wavy arpeggio sign in front of a chord and gently rolls the notes from bottom to top during playback while keeping the chord neatly aligned on the beat.
- Quantize Tools (Shortcut 'Q'): Snaps selected notes cleanly onto Quarter, 8th, 16th, 32nd, 64th, Triplet, or Swing beats.

## 11.4 Divisio Vocum: Spreading a Piano or Chord Sketch Across Your Orchestra

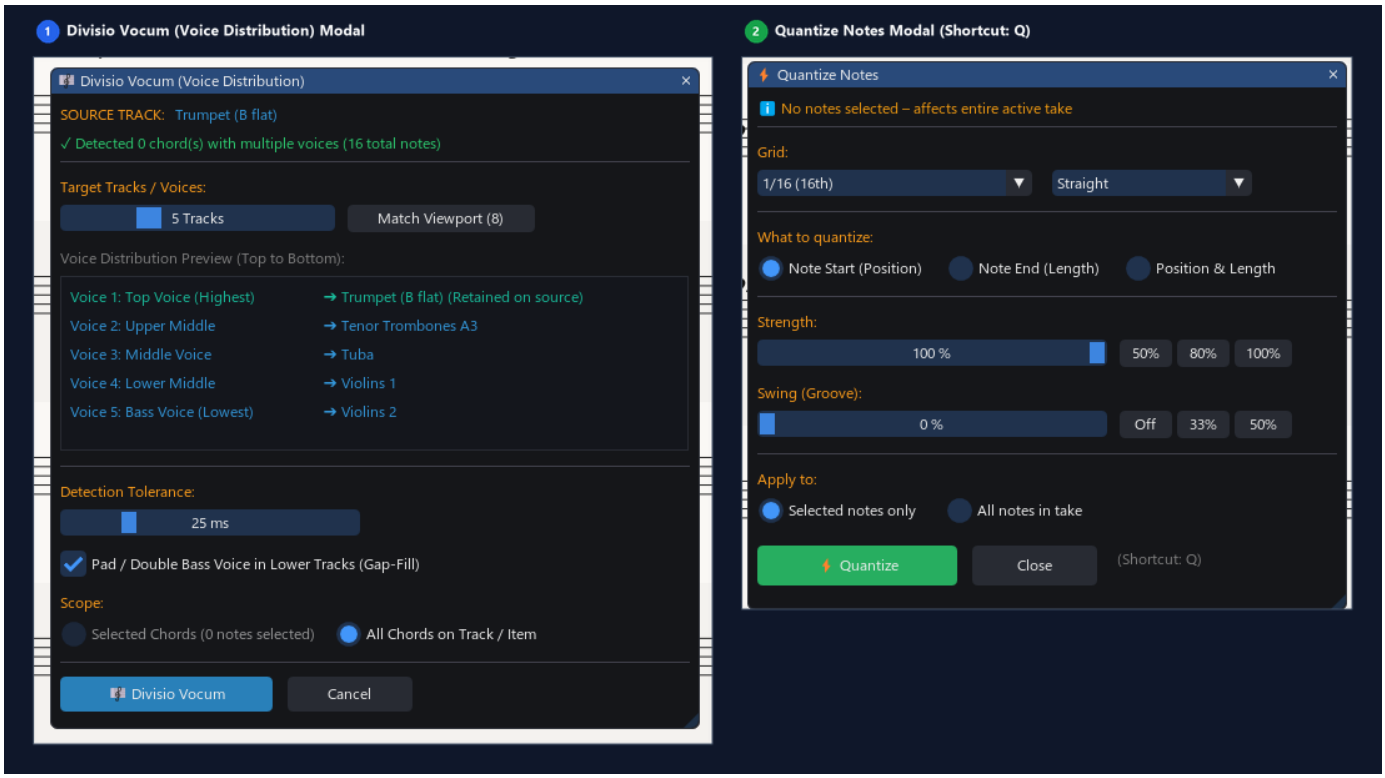



*Figure 11.2: Divisio Vocum (Voice Distribution) Modal (1) and Quantize Notes Dialog (Shortcut: Q) (2).*

What Divisio Vocum Does for You:
Many composers love to sketch chords with two hands on a single track (or on the first Violin track) and then spread those notes out across the string or brass section. Divisio Vocum takes the chords on your top track and automatically hands out each note-from highest to lowest-to the instrument tracks below it (for example: Violin 1 gets the top note, Violin 2 gets the second note, Viola gets the middle note, Cello gets the fourth note, and Double Bass gets the bottom note).

How to Open Divisio Vocum:

- From the Tools Panel: Open the 'Tools' panel on the right and click 'Divisio Vocum...'.
- By Right-Clicking the Score: Right-click any selected chord or bar on the page and choose 'Divisio Vocum...'.

Simple Controls Inside the Window:

- Target Tracks / Voices Slider (2 to 16): Choose how many instrument tracks will receive notes.
- 'Match Viewport' Button: Click this button to automatically match the exact number of instrument tracks currently visible on your screen.
- Live Routing Preview List: Shows you right away which track will receive the Highest note (Voice 1), Middle notes, and Lowest Bass note before you click Apply.
- Chord Detection Tolerance (5 ms to 100 ms): Even if you played your chords live on a MIDI keyboard and your fingers didn't land at the exact same millisecond, this slider groups the notes together as one chord.
- Pad / Double Bass Voice in Lower Tracks: If you play a 3-note chord across 5 string tracks, checking this box automatically doubles the lowest bass note in the Cello and Double Bass so every instrument has a note to play.
- Keeps Your Articulations Intact: If your chords had Staccato or Accent marks, each destination instrument automatically gets the matching Staccato or Accent sound from its own sound bank-and you can undo the whole arrangement anytime with Ctrl+Z.

## 11.5 Divisi: Splitting Multi-Voice Staves onto Cloned Instrument Tracks

What Is the Difference Between Divisio Vocum and Divisi?

- 'Divisio Vocum' spreads chords across DIFFERENT instrument tracks that already exist in your project (like Violin -> Viola -> Cello).
- 'Divisi' is for when a single section splits into parts (for example, 'Violins I divisi a 2' or 'Horns divisi a 3'). When you have Voice 1 and Voice 2 on one staff, Divisi keeps Voice 1 on your original track and automatically clones your REAPER track (including its loaded Kontakt/VSTi instrument and Reaticulate sound bank) once for each extra voice!

Why This Is So Useful:
Many solo legato sample patches can only play one note at a time-if you play two notes at once on the same track, one note gets cut off. With Divisi, Voice 2 automatically gets its own cloned track ('Track Name (Divisi 2)') with the exact same instrument loaded, so both legato voices sing out together!

How to Use Divisi:

- One-Click Right-Click Menu: Right-click on the staff and choose 'Divisi (Split Voices to Duplicated Tracks)'.
- Preview Window: Choose 'Divisi (Options / Preview)...' in the right-click menu or Tools panel to see how many notes are in each voice before splitting.

## 11.6 Filter Irregular Notes (Cleaning Up Live-Played Timing)

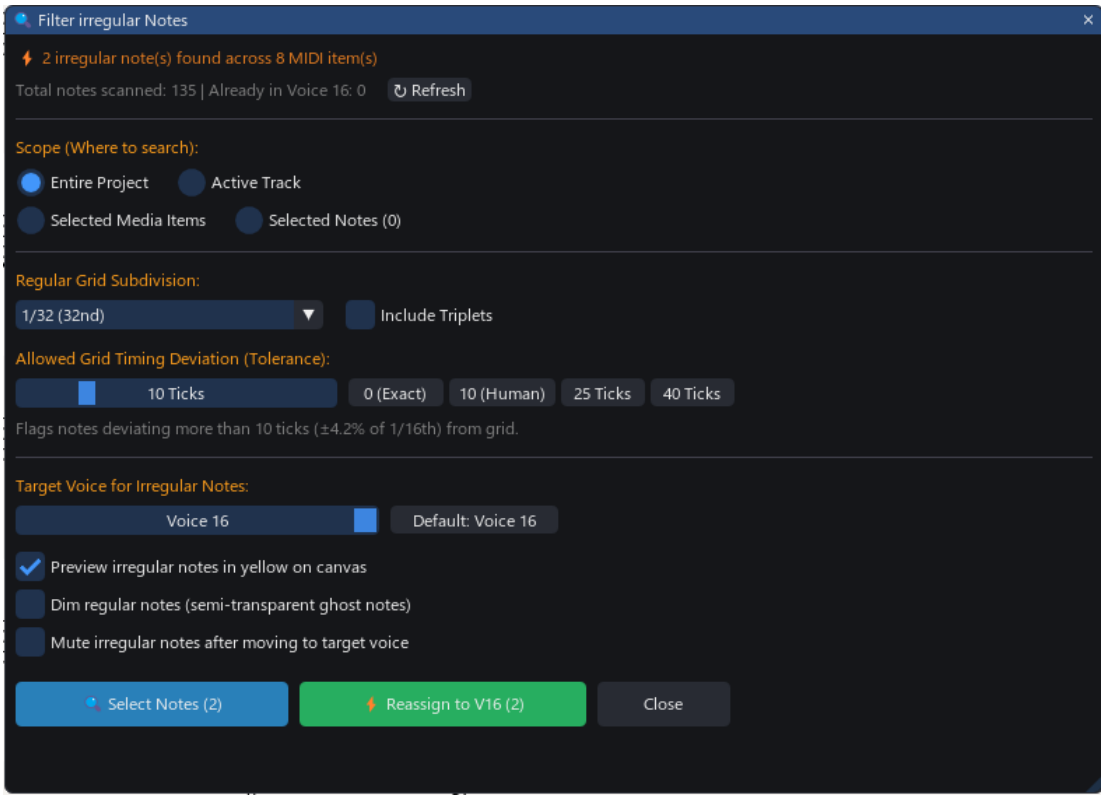



*Figure 11.3: Filter Irregular Notes Modal - Scope Selection, Regular Grid Subdivision, Tick Deviation Tolerance, Target Voice 16 Routing, and Live Yellow Canvas Preview.*

What This Tool Does:
When you record live without quantizing-or import a humanized MIDI file-some notes might sit slightly off the beat, creating extra rests or complex ties on the page. Filter Irregular Notes highlights those off-beat notes in bright yellow right on your score so you can either select them or move them to a background voice (like Voice 16).

Controls in the Window:

- Tolerance Slider: Slide left to catch even tiny timing slips, or slide right to only catch notes that are far off the beat.
- Include Triplets Checkbox: Recognizes triplet rhythms as valid on-beat notes.
- Target Voice Selector (Voice 1 to 16): Picks which voice off-beat notes will be moved into.
- Where to Look (Scope): Choose Entire Project, Active Track, Selected Clips, or Selected Notes Only.
- Two Action Buttons: Click 'Select Notes' to highlight the off-beat notes on the page, or click 'Reassign to Voice' to move them cleanly into your chosen background voice.

## 11.7 Adding Free Text Notes to the Page (Ctrl+T)

Press Ctrl+T (or choose the Text tool in the left sidebar) to place a text label anywhere on your score-great for performance notes like 'warm and expressive', 'mute on', or stage cues. Double-click any text item on the page to edit the words, font size, bold, or italic style.

## 11.8 Melodic Pathfinder: Intelligent Film-Score, Guitar/Bass & Counter-Melody Composer (v3.0)

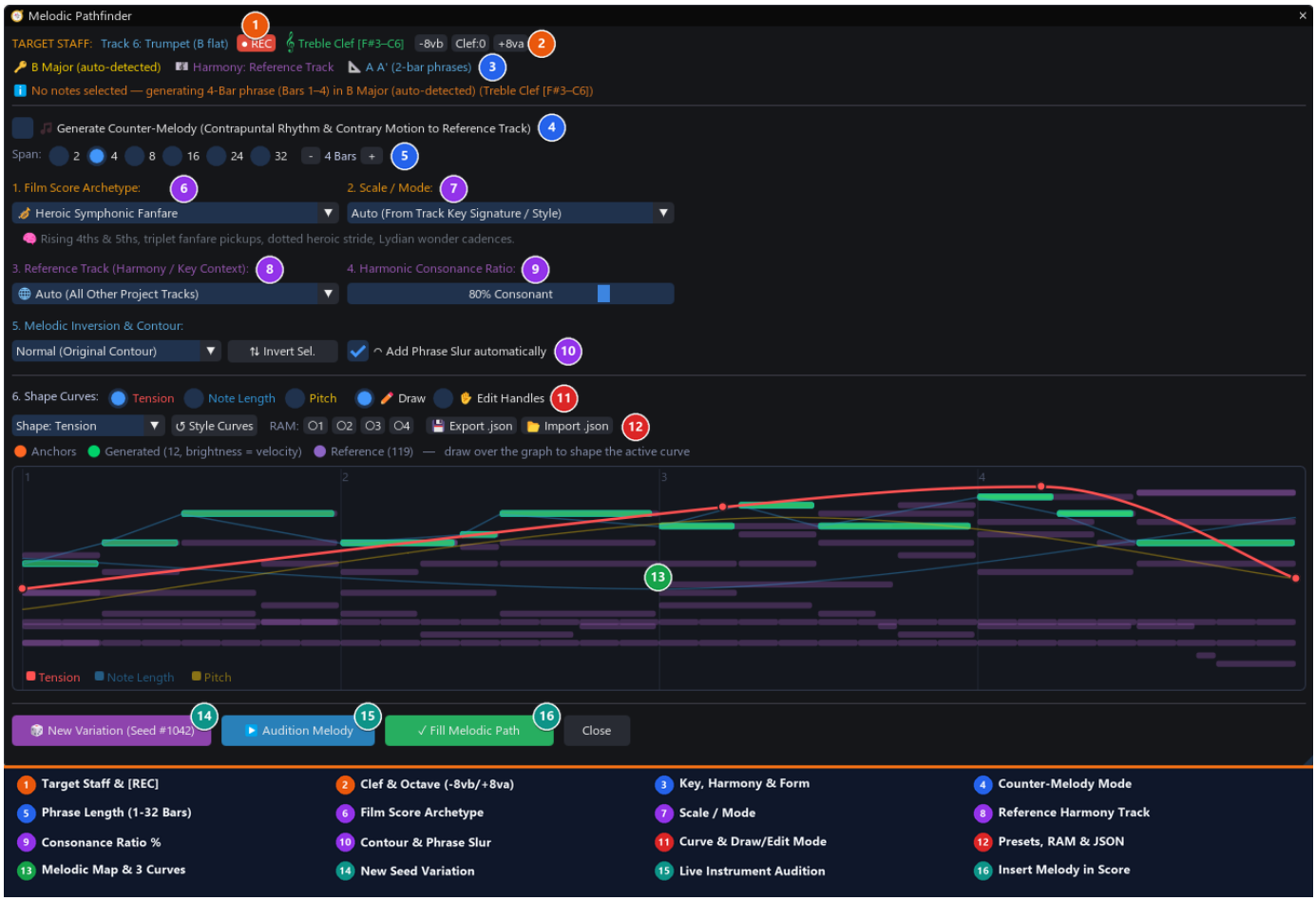



*Figure 11.4: The Melodic Pathfinder Window - Numbered Controls 1 through 16 (explained step by step below).*

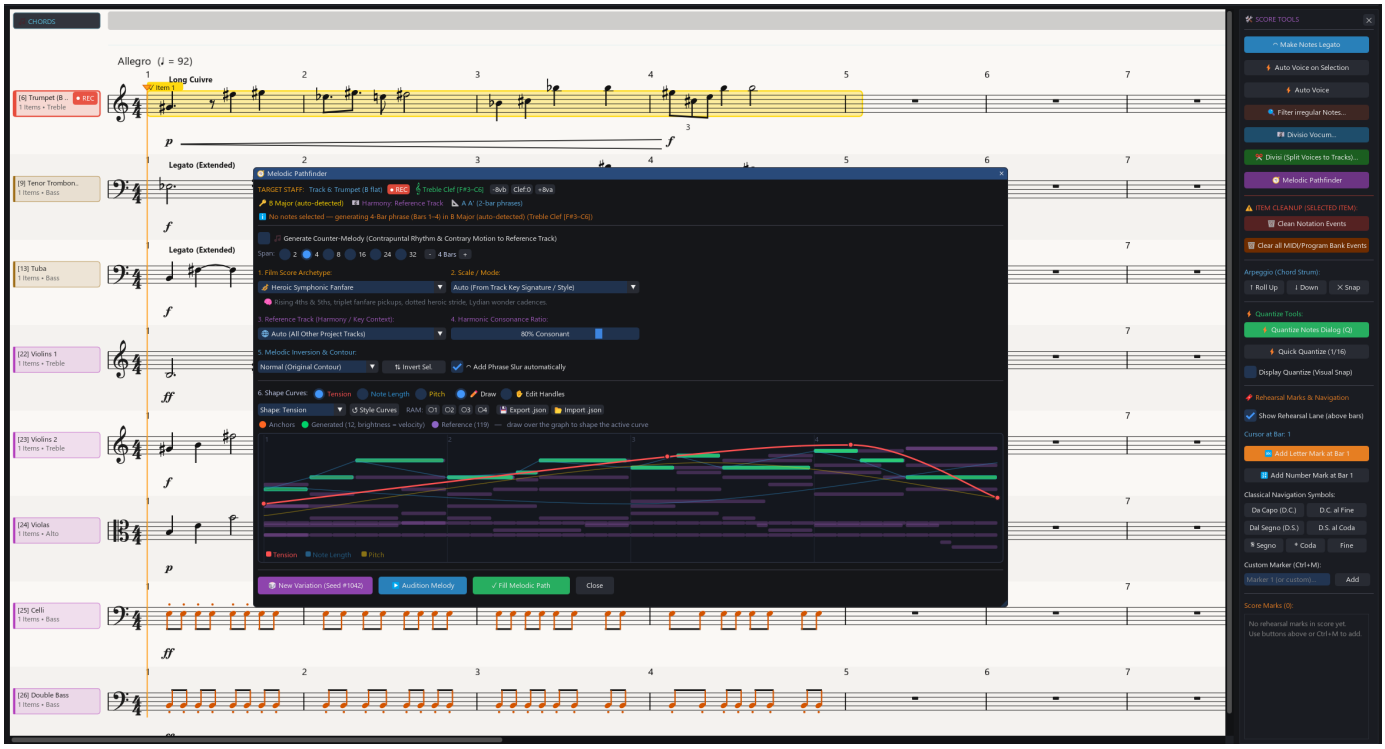



*Figure 11.5: Melodic Pathfinder Operating Live Over a Multi-Staff Orchestral Score Alongside the Score Tools Drawer.*

What Melodic Pathfinder Does for You:

Melodic Pathfinder is your musical co-composer inside REAPER-Notator. Whether you have sketched a few important notes and want a natural melody to connect them, want to grow a single starting note into a complete 4- to 32-bar film theme, or need an expressive counter-melody that weaves around another instrument in your orchestra, Melodic Pathfinder creates musical phrases that fit your track's clef, playable instrument range, key signature, and harmony.

How to Open Melodic Pathfinder:

- From the Score Canvas: Right-click anywhere on a staff (with 0, 1, or 2+ notes selected) and choose 'Melodic Pathfinder (Fill Between Key Notes)...'.
- From the Tools Drawer: Open the 'Tools' panel on the right side of the window and click 'Melodic Pathfinder (Fill Key Notes)...'.

The Three Ways to Use Melodic Pathfinder (Automatic Selection Detection):

- Workflow A - Blend Selected Anchor Notes (Select 2 or More Notes): Place two or more landmark notes in your score-for example, a starting note in Bar 1, a dramatic high peak in Bar 3, and a final cadence note in Bar 4-and select them before opening Melodic Pathfinder. Your selected notes remain 100% untouched (shown in Orange on the preview map), and Melodic Pathfinder composes the missing musical bridge connecting each selected note to the next.
- Workflow B - Grow a Theme from a Single Starting Note (Select 1 Note): Select just one note on the staff to serve as your starting note ('Seed Note') and choose a length from 2 up to 32 bars. Melodic Pathfinder develops a complete multi-bar theme starting from that exact note, complete with recurring motifs, natural breathing rests, and musical phrase endings.
- Workflow C - Create a Free Phrase or Counter-Melody (Select 0 Notes): Simply place the Edit Cursor where you want music to begin without selecting any notes. Melodic Pathfinder automatically detects the key and harmony from your other project tracks and writes a complete phrase-or, when 'Generate Counter-Melody' is checked, writes a contrapuntal line that moves in the opposite pitch direction and fills in the pauses of your main melody.

Numbered Guide to Every Button & Control (Matching Numbers 1 to 16 in Figure 11.4):

- [1] Target Staff & [REC] Ready Indicator: Shows which instrument track will receive the melody (e.g. 'Track 6: Trumpet (B flat)'). While the Melodic Pathfinder window is open, this track is automatically pre-armed-shown by the red '[REC]' badge both here and on the track's staff header in the score-so that clicking 'Audition Melody' plays the very first note cleanly without ever cutting off the attack. As soon as you close the window, the track automatically returns to its previous state.
- [2] Clef Range & Octave Shift Buttons (-8vb / Clef:0 / +8va): Displays your track's active clef and the natural playable range of your instrument (e.g. 'Treble Clef [F#3-C6]'). Click '-8vb' to shift the generated melody one octave lower, '+8va' to shift it one octave higher, or 'Clef:0' to return to the natural center register.
- [3] Automatic Key, Harmony Source & Phrase Form Display: Shows the musical context detected from your project at a glance:
- Key (e.g. 'B Major (auto-detected)'): Automatically detected from your track's Key Signature, Chord Lane, or surrounding tracks.

- Harmony Source (e.g. 'Harmony: Reference Track'): Shows where the underlying chord changes come from so melody notes lock cleanly onto chord tones on every half-bar.
- Phrase Form (e.g. 'A A' (2-bar phrases)' or 'A A' B A'"): Shows how the melody is structured into statement, variation, contrast bridge, and return-just like a real soundtrack theme.
- [4] Generate Counter-Melody Checkbox: Check this box when you already have a main melody on another track and want a secondary line (such as French Horn or Cello answering the Violins). In Counter-Melody mode, the new melody moves in contrary motion (stepping down when the main melody rises, and rising when the main melody falls) and plays active rhythmic figures whenever the main melody holds a long note or rests.
- [5] Phrase Length Selector (Span: 2 / 4 / 8 / 16 / 24 / 32 & - / + Step Buttons): When 0 or 1 note is selected, click a quick preset (2, 4, 8, 16, 24, or 32 bars) or use the '-' and '+' buttons to set any exact length from 1 to 32 bars. (When 2 or more Anchor Notes are selected, the span automatically matches the distance between your first and last selected note.)
- [6] 1. Film Score Archetype Dropdown: Choose from 8 descriptive soundtrack styles (listed in detail below). Each archetype brings its own characteristic intervals, rhythm feel, motif memory, and default phrasing curves.
- [7] 2. Scale / Mode Dropdown: Leave on 'Auto (From Track Key Signature / Style)' for automatic key matching, or manually choose Major (Ionian), Natural Minor (Aeolian), Harmonic Minor, Dorian, Phrygian, Lydian, Mixolydian, Minor Pentatonic, or Chromatic.
- [8] 3. Reference Track (Harmony / Key Context): Choose which track Melodic Pathfinder listens to for harmony and counterpoint. 'Auto (All Other Project Tracks)' listens to your whole arrangement, or you can pick a specific track (e.g. Piano or Strings) or 'None'.
- [9] 4. Harmonic Consonance Ratio Slider (0% to 100%): Controls how closely the melody hugs the chords of your Reference Track. Set to 75%-100% for warm, consonant harmony (chord tones, 3rds, 5ths, and 6ths), around 50% for expressive passing notes, or 0%-25% for dark, gritty thriller tension.
- [10] 5. Melodic Inversion & Contour, 'Invert Sel.' Button & Phrase Slur Checkbox:
- Contour Dropdown: Keep the melody 'Normal (Original Contour)', flip its intervals upside-down ('Inversion'), reverse it in time ('Retrograde'), or combine both ('Retrograde-Inversion').
- 'Invert Sel.' Button: Immediately flips any currently selected notes on the score canvas upside-down around the first selected note.
- 'Add Phrase Slur automatically' Checkbox: Automatically places a smooth legato slur over the generated phrase so Reaticulate triggers connected legato transitions on your instrument.
- [11] 6. Shape Curves Selector (Tension / Note Length / Pitch) & Tool Mode (Draw / Edit Handles):
- Choose which of the three colored curves you want to edit: 'Tension' (Red), 'Note Length' (Blue), or 'Pitch' (Gold).
- 'Draw' Mode (Pencil): Click and drag directly across the note map to sketch the curve freehand. The moment you release the mouse button, your hand-drawn line is automatically smoothed into a clean Bezier curve with control handles.
- 'Edit Handles' Mode (Hand): Drag the round anchor points or bend the tangent handles to fine-tune the curve shape with precision. Double-click on the graph in Edit mode to add or remove a control point.
- [12] Curve Presets, Reset, 4 RAM Slots & JSON Export/Import:
- 'Shape: ...' Dropdown: Apply ready-made dramatic shapes (such as Arch, Late Climax, Swell, Ramp Up, Decay, or Wave) to the active curve.
- 'Style Curves' Button: Resets all three curves back to the natural default shape of the chosen Film Score Archetype.
- 'RAM: 1 2 3 4' Slots: Left-click an empty slot to save your current 3 curves in memory; left-click a filled slot to recall it; right-click a filled slot to overwrite it.
- 'Export .json' & 'Import .json': Save your favorite curve setups to disk as a .json file or load them in any other project.
- [13] Interactive Melodic Map & Bezier Curve Canvas:
- Orange Bars: Your selected Anchor Notes (locked in place and preserved 100%).
- Green Bars: The generated melody notes. Brighter green bars indicate higher note velocity (louder dynamics); darker green bars indicate softer dynamics.
- Purple Bars: Harmony notes from your Reference Track so you can see how your new melody sits inside the accompaniment.
- Colored Curves: The Red (Tension), Blue (Note Length), and Gold (Pitch) curves drawn right over the measures.
- [14] 'New Variation (Seed #...)' Button: Click the dice button to explore a fresh melodic variation while keeping all your chosen style, harmony, and curve shapes.
- [15] 'Audition Melody' (or 'Stop Audition') Button: Plays the generated melody live through your track's virtual instrument-including its active Reaticulate articulation and dynamic velocity shading-so you can hear the result before inserting it.
- [16] 'Fill Melodic Path' (or 'Insert Counter-Melody') & 'Close' Buttons: Inserts the melody into your score in a single Undo step (Ctrl+Z) and automatically spells all sharps and flats to match your Key Signature.

How the 3 Interactive Shape Curves Work Together:

- 1. Tension Curve (Red) - Controls Drama, Velocity & Leaps: When you draw the Red Tension curve low, the melody plays softly (pp to mp velocity) using gentle, calm steps. As you draw the Tension curve higher, note velocities automatically rise (mf to ff), intervals open up into wider heroic leaps, and long notes lean into expressive emotional suspensions before resolving.

- 2. Note Length Curve (Blue) - Controls Rhythmic Speed: Drawing the Blue curve low creates faster, more active rhythmic motion (8th and 16th notes); drawing it high creates broad, singing sustained notes (half and whole notes).
- 3. Pitch Curve (Gold) - Controls Melodic Height: Guides where the melody climbs toward a high register climax and where it settles down into a lower register cadence. Steep upward climbs in the Pitch curve also naturally boost musical tension.

The 8 Descriptive Film Score Archetypes:

- 1. Heroic Symphonic Fanfare: Bold rising 4ths and 5ths, triplet fanfare pickups, dotted heroic stride, and bright uplifting cadences-ideal for brass, horns, and main title themes.
- 2. Intimate Elegiac Strings: Singing stepwise cantabile lines, expressive 6th leaps, sighing downward resolutions, and emotional suspension notes-ideal for solo violin, cello, oboe, or lyrical string sections.
- 3. Minimalist Ostinato Pulse: Hypnotic repeating motifs that circle around a central pivot note while gradually building tension-ideal for modern drama, piano/mallet pulses, and driving string textures.
- 4. Pastoral Folksong & Wonder: Warm, open modal intervals with natural breathing rests and gentle cadences-ideal for flute, clarinet, horn, and sweeping landscape scores.
- 5. Gothic Fairy-Tale Waltz: Playful chromatic neighbor notes, music-box leaps, and dancing rhythmic lilt-ideal for celesta, pizzicato strings, harp, bassoon, and fantasy cues.
- 6. Poetic Suspense & Noir: Restrained, mysterious phrases with dark modal colors and long held notes answered by restless rhythmic figures-ideal for muted trumpet, alto flute, solo viola, or noir piano.
- 7. Epic Ancient Brass & Choir: Monumental, broad themes built on powerful 4ths, 5ths, and octaves with solemn weight-ideal for low brass, horns, choir, and historical/mythic scores.
- 8. Romantic Contrapuntal Cantabile: Flowing, song-like classical lines with balanced arches and graceful motivic development-ideal for woodwind solos, chamber strings, and expressive counter-melodies.

Guitar & Bass Mode + Custom Key Override (v3.0):

- Switch Between [Orchestral / Melodic] and [Guitar / Bass] Tabs: At the top of Melodic Pathfinder, switch to the Guitar / Bass tab to compose idiomatic fretted parts for 6-String Guitar, 5-String Guitar, 4-String Guitar, or 4/5-String Bass.
- Four Guitar Playing Modes: Choose 'Melody (Single-Note Lead)', 'Fingerpicking (Polyphonic Arpeggio)', 'Strum (Rhythmic Chords)', or 'Mixed (Strum + Picking)' across 8 acoustic, pop, rock, and cinematic guitar styles.
- Global or Bar-by-Bar Key Override: Choose 'Auto (From Project)', lock the entire phrase to a specific Global Key (e.g. 'D Minor'), or set custom Bar-by-Bar key changes across your phrase.

## 11.9 Drawing Freehand Notes & Markings on Your Score ('Layers' Panel)

Want to circle a bar with a red pen, draw an arrow, or jot down rehearsal markings by hand just like on a paper score?

- Open the '[Layers] Layers' Panel on the Right: Create as many drawing layers as you like for each instrument track, and show or hide them with a click.
- Smooth Pen & Eraser Tools: Pick the Pen tool, choose your brush thickness and one of the 12 colors, and draw right over the staves. Switch to the Eraser anytime to wipe away strokes.
- Automatically Saved With Your Project: Your hand-drawn notes are saved automatically as transparent images alongside your project, and you can double-click any layer name to rename it.

## 11.10 Circle of Fifths & Chord Progression Builder

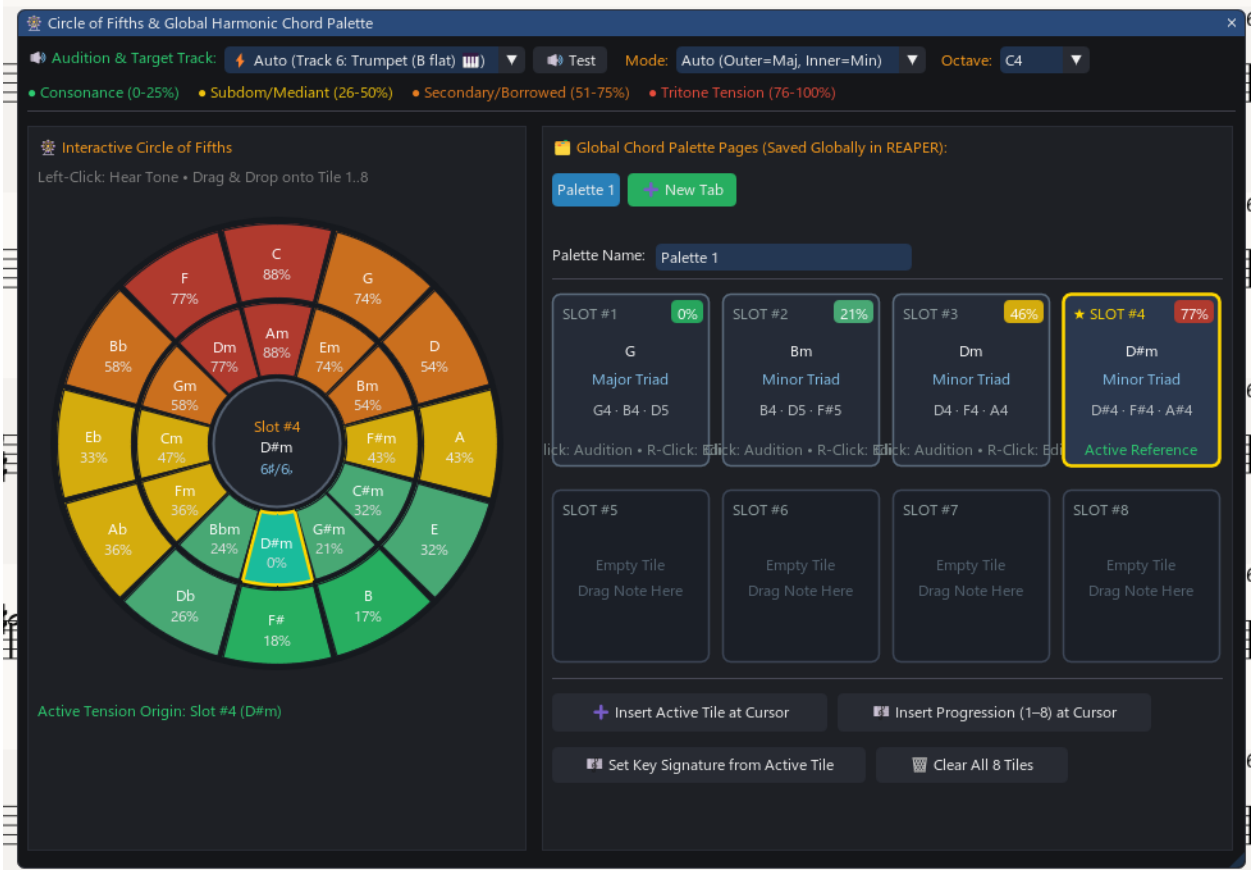



*Figure 11.7: Circle of Fifths & Global Harmonic Chord Palette Modal - Interactive 24-Sector Harmonic Wheel, Consonance/Tension Coloring, Multi-Tab Palette Pages, and 8 Progression Slots.*

Even if you haven't studied music theory, the Circle of Fifths window makes finding great-sounding chords and building chord progressions effortless:

- How to Open It: Click the 'Circle of Fifths' button right below 'Step Input' in the left sidebar.
- Click Any Key on the Wheel to Hear It: The outer ring shows the 12 Major keys (C, G, D, A, E, B, F#, Db, Ab, Eb, Bb, F) and the inner ring shows their matching Minor keys (Am, Em, Bm, etc.). Click any slice of the wheel to hear that chord played live through your chosen instrument track!
- Pick Rich Chord Flavors & Voicings: Use the buttons at the top to turn simple chords into 7th, 9th, sus4, add9, 6th, or dim7 chords, flip inversions (Root, 1st Inversion, 2nd Inversion, or wide Orchestral Spread), and pick the octave.
- Build Your Chord Progression in the 8 Slots: Drag any chord from the wheel into Slots 1 through 8 on the right (or double-click a slice on the wheel). When you click a chord slot, the wheel lights up in color to show you which next chords will sound smooth and harmonious (Green and Blue) and which chords will create dramatic tension (Amber and Red)!
- One-Click Insert into Your Score: Click 'Insert' on a single chord slot-or 'Insert All Tiles at Cursor'-to write your entire chord progression straight into your score (and if you have multiple tracks armed with Auto-Divisi turned on, it will even spread the chords across your orchestra automatically!).

## 11.11 Writing Vocal Lyrics & Live Sync with Synthesizer V Studio Pro 2

Writing Lyrics Under Vocal Staves:

- Dedicated Lyrics Line: Vocal tracks display a clean lyrics line right below the staff, complete with centered hyphens ('-') between syllables and hold lines ('_') under notes held across multiple pitches.
- Double-Click Any Note to Type Lyrics (or Press Ctrl+L): Double-click a note in the lyrics lane and type your words! If you type a whole phrase separated by spaces or hyphens (for example: `Hel-lo world _`), REAPER-Notator automatically spreads the syllables across the following notes for you. Press Tab to jump to the next note, or Shift+Tab to go back.
- Easy Lyric Erasing: Delete the text on a note or click 'Clear Note Lyric' / 'Clear Track' in the Lyrics panel to remove lyrics cleanly.
- One-Click 'Duplicate Lyrics Track' Button: Click 'Duplicate Lyrics Track' in the Lyrics panel to create a complete 1:1 duplicate of your vocal track-including all MIDI notes, sung lyrics, text passages, dynamics, hairpins, and slurs-ideal for creating vocal doubles or harmony backing tracks.

Live Two-Way Bridge to Synthesizer V Studio 2 Plugin (AI Singers):

- Automatic Connection to Your Vocal Tracks: If you have the 'Synthesizer V Studio 2 Plugin' (VST3i) loaded on a vocal track in REAPER, REAPER-Notator links that track directly 1-to-1 with its Synthesizer V plugin window (while ignoring drum and non-vocal tracks).
- Sends Notes, Lyrics & Volume Dynamics Together: Your melody notes, your typed lyrics, and even your score volume markings (pp to ff, accents like sfz, and crescendo/diminuendo hairpins) are sent straight into Synthesizer V-preserving your chosen AI singer voice (such as SOLARIA II) and vocal style so your virtual singer performs your sheet music immediately!

# Chapter 12: Opening & Saving Sheet Music (MusicXML 4.0 Import & Export)

## 12.1 Sharing Sheet Music With Other Programs

MusicXML is the standard file format used by virtually all sheet music programs and online score libraries. REAPER-Notator can both open (Import) and save (Export) MusicXML 4.0 files directly.

## 12.2 Importing a MusicXML Score into REAPER

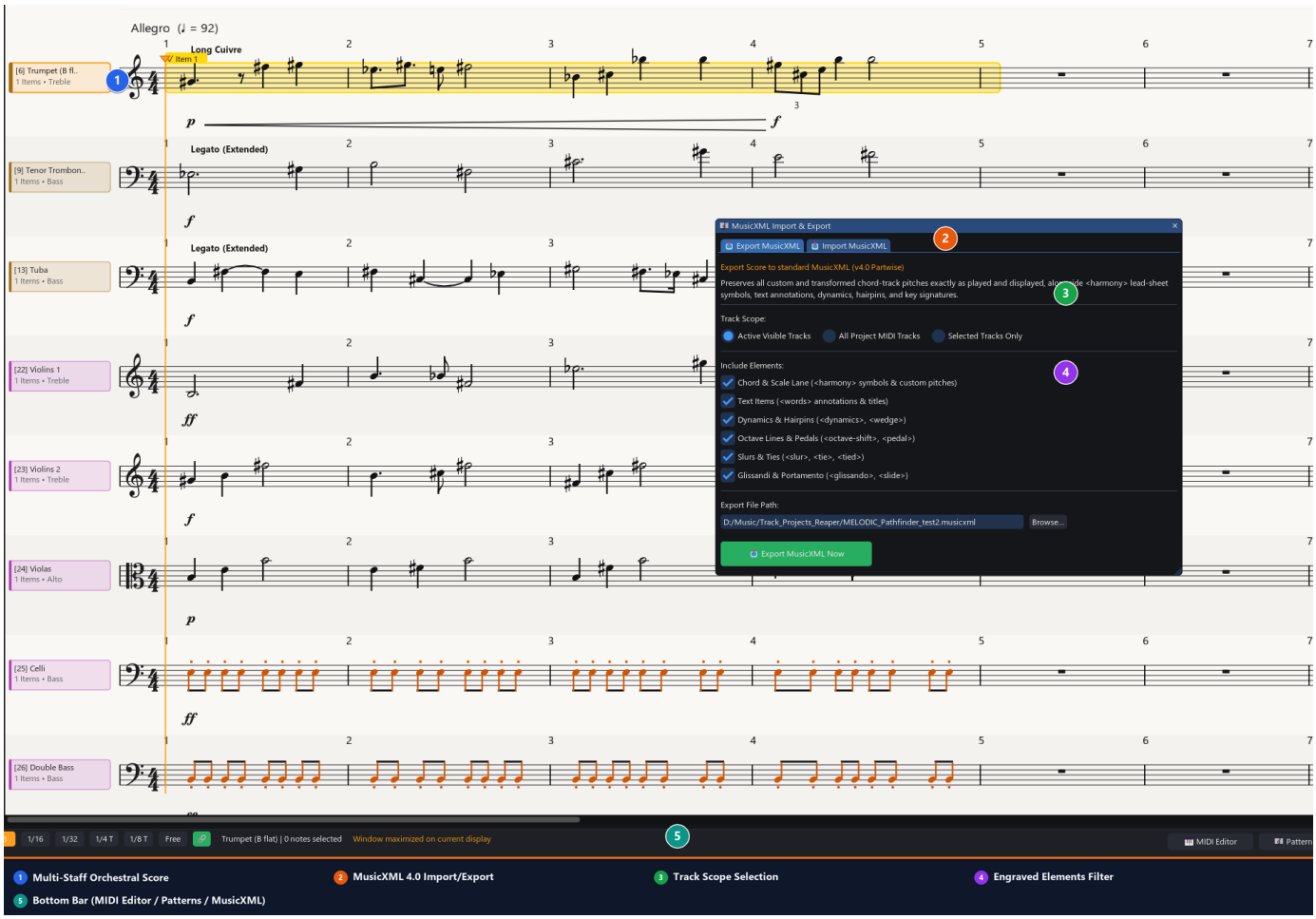



*Figure 12.1: Multi-Staff Orchestral Score Canvas & MusicXML 4.0 Import/Export Modal.*

How to Import Sheet Music:

- Click 'Import MusicXML': Select any `.musicxml` or `.xml` file on your computer. REAPER-Notator automatically creates the instrument tracks in REAPER and imports all measures, time signatures, key signatures, notes, dynamics, lyrics, slurs, ties, and tempo changes.
- Automatic Time Signature & Articulation Matching: Mid-piece time signature changes are automatically added to REAPER's main timeline bar ruler, and markings like Staccato or Pizzicato are automatically matched to your loaded Reaticulate sound banks.
- Neat Track Layout: Imported tracks are neatly sized in REAPER so even a large orchestral score stays tidy.

## 12.3 Exporting Your Project as MusicXML

Click 'Export MusicXML' to save your REAPER composition as a standard `.musicxml` file that you can send to live musicians, conductors, or other notation software.

# Chapter 13: Copying & Pasting Notes and Markings (Ctrl+C / Ctrl+V)

## 13.1 Smart Copy & Paste Without Messy Overlaps

Copying and pasting in REAPER-Notator works just the way you expect:

- Copy Only What You Select (Ctrl+C): When you select notes and press Ctrl+C, REAPER-Notator copies those notes and their attached articulations (like staccato dots or accents) without accidentally grabbing unselected dynamics or tempo marks.
- Clean Paste Without Double-Notes (Ctrl+V): Place the Edit Cursor where you want to paste and press Ctrl+V. Any old notes sitting in that time range are cleanly replaced by your pasted phrase so you never end up with messy double-stacked notes, and your instrument sound switches update automatically.
- Copy & Paste Markings Too: You can also select, copy, and paste dynamics, hairpins, and pedal lines on their own.

## 13.2 Pasting Across Different Instrument Tracks

Want to copy a melody from the Flute to the Oboe or Violin? Simply select the notes on the first track, press Ctrl+C, click on the new instrument's staff, and press Ctrl+V-the phrase is pasted at the exact same beat position on the new instrument.

# Chapter 14: Smooth Playback & Performance on Large Scores

## 14.1 Zero CPU Waste When Your Score Is Standing Still

When playback is stopped and you aren't moving notes around, REAPER-Notator automatically reuses your already-calculated page layout instead of re-reading your project over and over. This keeps your computer quiet and saves CPU power for your audio plugins and virtual instruments.

## 14.2 Graphics Card (GPU) Drawing

All staves, noteheads, beams, and slurs are drawn directly by your computer's graphics card, keeping zooming and scrolling crisp and responsive.

## 14.3 Smooth 60+ FPS Playback Even With 30+ Instrument Tracks

While your music is playing back, REAPER-Notator focuses purely on scrolling the page and moving the playhead smoothly-delivering 60+ frames per second even on large symphonic scores with dozens of instruments.

## 14.4 Only Drawing What Is Currently on Your Screen

If your piece is 200 bars long, REAPER-Notator only draws the bars and instruments that are actually visible inside your window right now, so long pieces stay just as fast as short 4-bar sketches.

## 14.5 Jitter-Free Auto-Scroll

When 'Follow Playhead' (Auto-Scroll) is turned on, the score and the playhead move together in lockstep without shaking, and when you stop playback you can freely scroll anywhere in your score to edit.

## 14.6 Instant Slurs, Ties & Time Signatures on Big Imported Scores

Even if you import a massive film or orchestral MusicXML score with hundreds of ties, slurs, and meter changes, REAPER-Notator organizes them in memory so scrolling and editing stay fast and lag-free.

# Chapter 15: Built-In Bottom MIDI Velocity & Controller Editor

## 15.1 Opening the Bottom MIDI Lane

How It Works:

- Click the Bottom Bar to Open/Close: At the very bottom of the REAPER-Notator window, click the MIDI / Velocity bar to slide open a built-in controller editor directly underneath your sheet music.
- Always Linked to Your Sheet Music: Every note on the staff lines up vertically with its velocity bar and controller curve below, so you can shape your sound while looking right at the notes.

## 15.2 Adjusting How Hard Notes Are Played (Velocity Lane)

How to Shape Note Velocities (1 to 127):

- Drag the Little Flag Handles Up or Down: Each note shows a vertical bar with a flag at the top-drag it up to make that note louder/stronger, or down to make it softer, while hearing the change live.
- Freehand Pencil Drawing: Switch to the Pencil tool to sweep across multiple notes and draw a smooth velocity curve in one motion.
- One-Click Volume Buttons: Click `pp` (32), `mp` (64), `mf` (80), `f` (96), or `ff` (112) to set selected notes to a clean starting volume.
- 'Ramp' & 'Humanize' Buttons: Click 'Ramp' to create a smooth fade-in or fade-out across selected notes, or click 'Humanize' to add subtle, natural player variations so repeated notes don't sound robotic.

## 15.3 Drawing Smooth Controller Curves (Modwheel CC1, Expression CC11, Pedal CC64 & More)

How to Edit Any MIDI Controller (CC 0 to 127):

- Pick Any Controller from the Dropdown: Switch between Modulation Wheel (CC 1), Breath (CC 2), Volume (CC 7), Pan (CC 10), Expression (CC 11), Sustain Pedal (CC 64), or any other controller.
- Draw Smooth Curves with the Pencil: Paint expressive swells directly in the lane.
- No Step-Stair Jumps: Uses smooth ramps and curves between points so your orchestral strings and winds swell seamlessly without zipper steps.
- Quick Level & Clear Buttons: Set the lane to 0, 32, 64, 96, or 127 in one click, or click 'Clear CC' to start fresh.

## 15.4 Automatic Lock Protection When Score Dynamics Are Active

How REAPER-Notator Protects Your Dynamic Markings:

- Automatic Padlock Icon: When your sheet music has active Dynamics or Hairpins shaping CC 1 (Modwheel) or CC 11 (Expression), the bottom lane displays a '[Locked] CC Locked (Dynamic Shaping)' badge so you don't accidentally scribble over your score dynamics.
- One-Click Unlock: Want to draw the curve completely by hand instead? Simply click 'Unlock (Enable Bypass)' to take full manual control.

# Chapter 16: Customizing Settings, Mouse Navigation & Colors

## 16.1 Choosing Your Screen Refresh Rate (FPS)

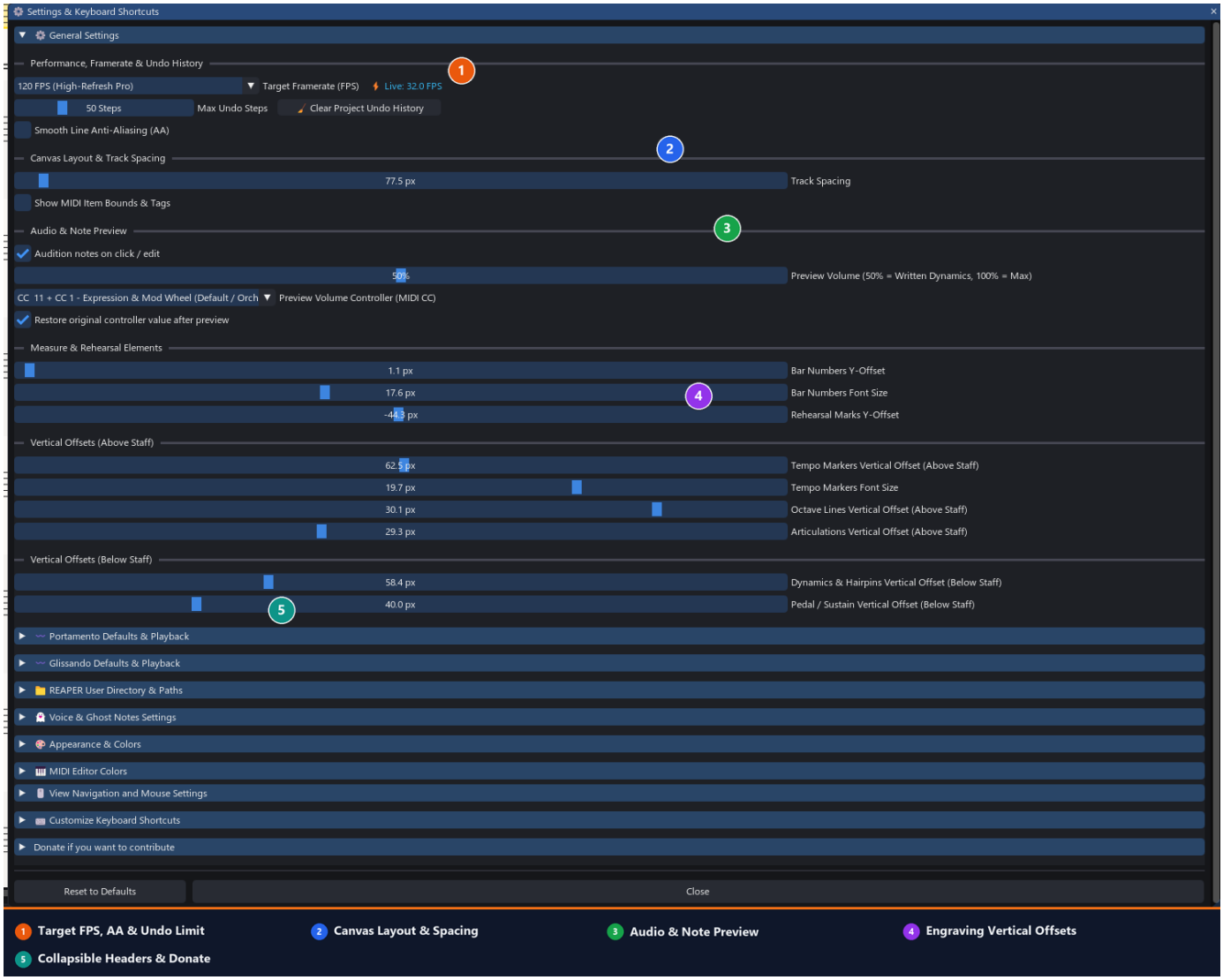



*Figure 16.1: Settings Modal Overview - Target FPS & Smooth Line AA (1), Canvas Layout & Spacing (2), Audio & Note Preview (3), Engraving Vertical Offsets (4), and Collapsible Category Headers including Donate Section (5).*

In Settings -> General Settings, you can choose how smoothly the window animates:

- Pick Your Frame Rate Preset: Choose 15 or 30 FPS to save battery on a laptop, 60 FPS (the standard smooth setting), or 90 / 120 / 144 FPS if you have a fast gaming/studio monitor.
- Built-In Smooth Timer: On Windows, REAPER normally limits script windows to around 30 frames per second. REAPER-Notator includes an automatic high-speed timer booster so your score scrolls at a silky-smooth 60 to 144 FPS while the window is open, and returns REAPER to normal the moment you close it.

## 16.2 Smooth Line Smoothing (Anti-Aliasing)

In Settings -> General Settings:

- Leave 'Smooth Line AA' Unchecked for Maximum Speed on huge 30+ track orchestral projects, or check it for extra-soft edges on slanted beams and hairpins. (Your musical notes, clefs, and text are always crystal clear in both modes.)

## 16.3 Undo History Limit & Freeing Up Memory

In Settings -> General Settings:

- Max Undo Steps Slider (Default: 50): Controls how many Undo steps are kept in memory.
- 'Clear Project Undo History' Button: Click this button anytime if you want to clear out old undo history and free up computer RAM on a long session.

## 16.4 Customizing Mouse Wheel Scrolling, Panning, Zooming & Glissando Defaults

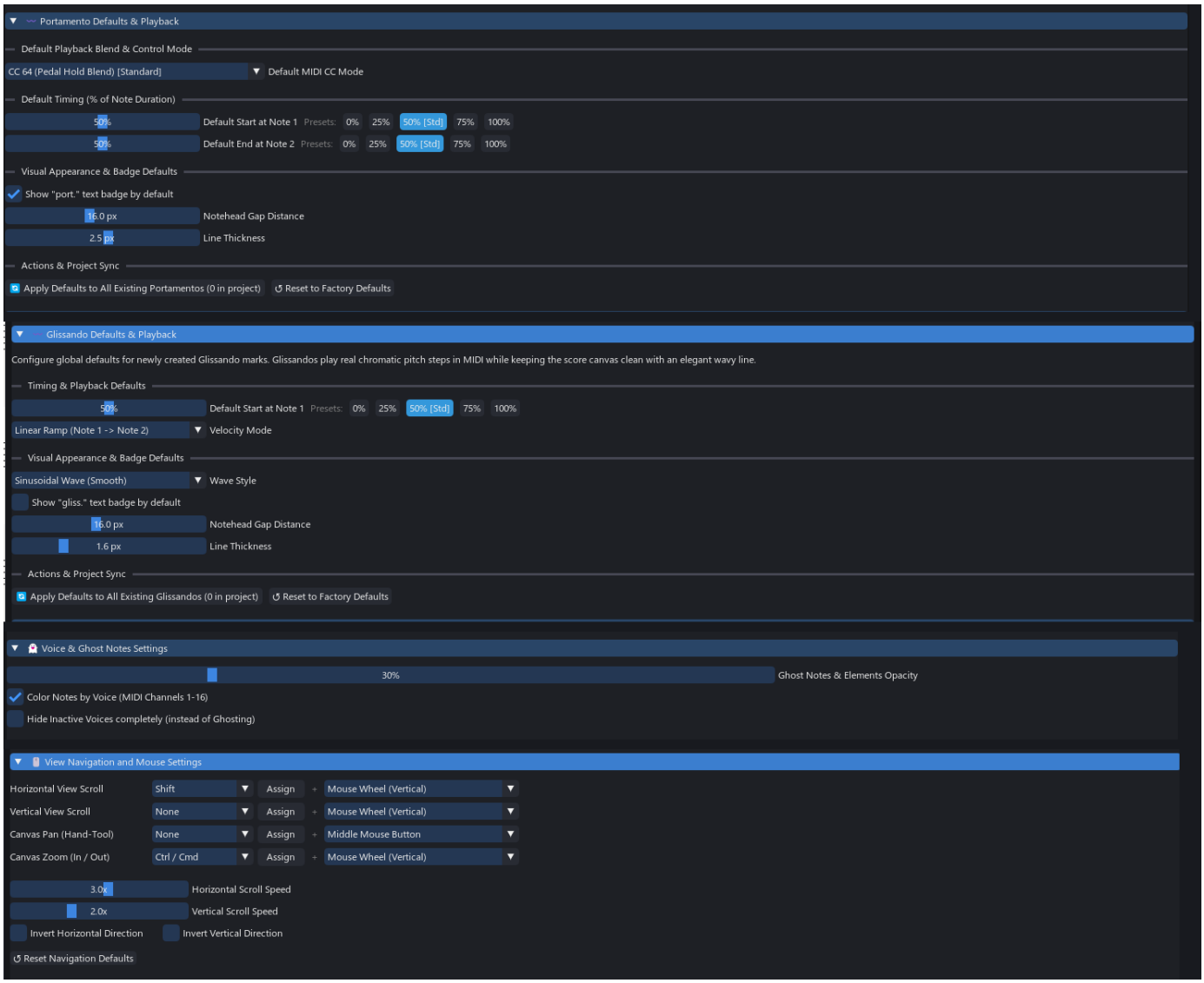



*Figure 16.2: Expanded Settings Panels - Portamento Defaults & Playback, Glissando Defaults & Playback, Voice & Ghost Notes Opacity, and View Navigation & Mouse Settings.*

Every musician has their own favorite way to scroll and zoom around a score. Open Settings -> View Navigation & Mouse Settings to make the mouse work your way:

- Customize Any Action: Choose which mouse button or wheel + key (Shift, Ctrl, Alt, Space, or None) controls Horizontal Scroll, Vertical Scroll, Hand-Tool Dragging (Pan), and Zooming.
- Scroll Speed & Direction: Adjust how fast the page scrolls (from 0.5x to 5.0x) or reverse the scroll direction with the 'Invert' checkboxes.
- 'Reset Navigation Defaults' Button: Returns all mouse controls to the standard setup in one click.

## 16.5 Changing Score & MIDI Editor Colors, UI Light/Dark Theme, Languages (EN/DE/FR) & Updates

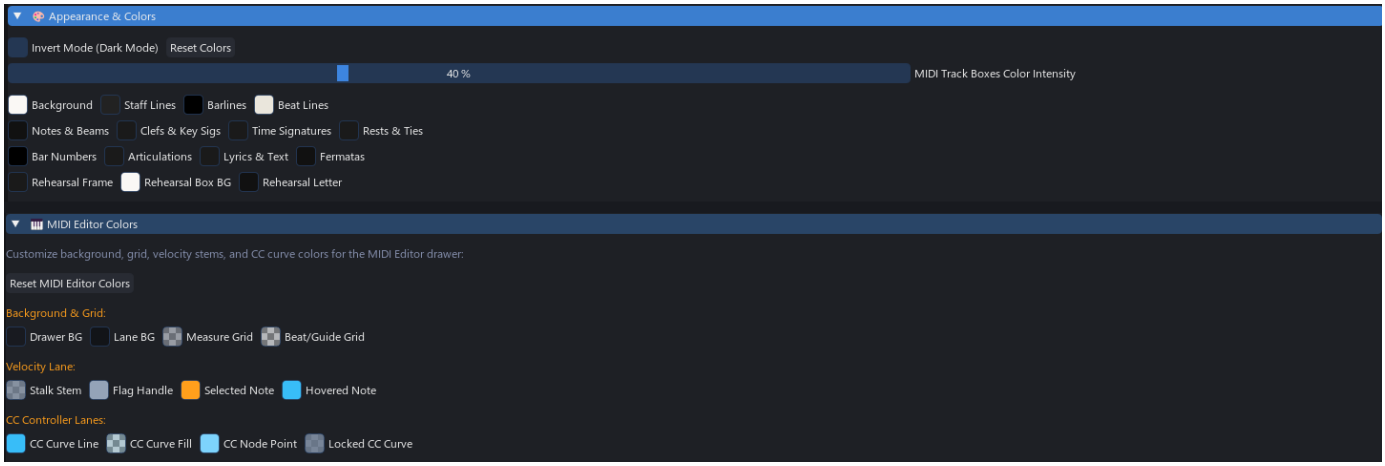



*Figure 16.3: Appearance & Colors (Dark Mode Inversion & Score Element Swatches) and MIDI Editor Colors.*

Open Settings -> General Settings or Appearance & Colors to personalize the look and language of REAPER-Notator:

- UI Theme (Dark UI / Light UI): Switch the entire application interface (toolbars, sidebars, drawers, and pop-up windows) between sleek Studio Dark UI and bright Daylight Light UI.
- Multi-Language Interface (English / Deutsch / Francais): Switch all menus, buttons, tooltips, and dialogs live between English (EN), German (DE), and French (FR) in Settings -> General Settings.
- Score Dark Mode Toggle: Switches the score page between classic light parchment paper and studio Dark Mode.
- Pick Custom Colors for Every Element: Click any color box to customize the colors of staves, barlines, notes, selection highlights, MIDI clip boxes, and velocity bars.
- Automatic Update Checker: In the background on startup, REAPER-Notator checks if a new version of REAPER-Notator or Cockos REAPER is available and displays a convenient green 'New Update Available' button in the top bar with 1-click ReaPack synchronization.

# Chapter 17: Keyboard Shortcuts & Quick Reference Table

## 17.1 Complete Keyboard Shortcut Cheat Sheet

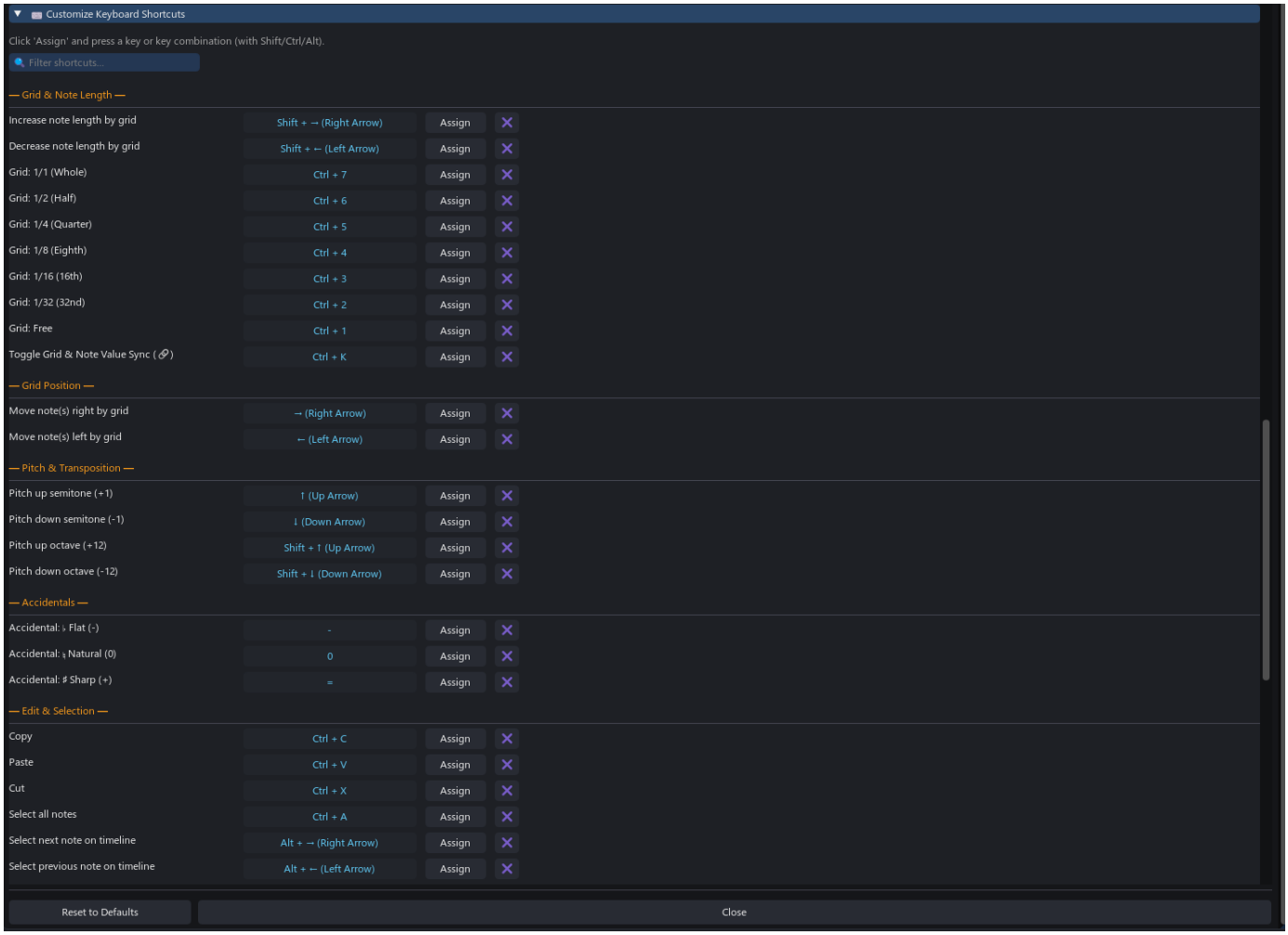



*Figure 17.1: Customize Keyboard Shortcuts Panel - Real-Time Shortcut Search Filter and 1-Click Key & Modifier Assignment.*

Handy reference table of default keyboard shortcuts (you can customize any of these in Settings -> Customize Keyboard Shortcuts):

| Key / Shortcut | Action | What It Does |
|---|---|---|
| :--- | :--- | :--- |
| Space | Play / Pause | Starts or pauses playback |
| Record Button (Left/Right C | Record / Record Settings | Left-click starts/stops recording; Right-click opens Recording Settings |
| Ctrl + M | Add Custom Marker | Drops a numbered Rehearsal Marker at the Playhead or Edit Cursor |
| F11 / Maximize Button | Maximize / Restore | Fills your monitor with the score or returns to a centered window |
| Delete / Backspace | Delete | Deletes selected notes, dynamics, slurs, ties, or markings |
| Esc | Clear Selection | Unselects everything |
| Up / Down Arrows | Move Pitch | Moves selected notes up or down by 1 half-step |
| Shift + Up / Down | Move Octave | Moves selected notes up or down by 1 full octave |
| Left / Right Arrows | Move in Time | Shifts selected notes earlier or later by 1 grid step |
| Shift + Left / Right | Change Note Length | Makes selected notes shorter or longer by 1 grid step |
| Alt + Left / Right | Step Through Notes | Jumps to the previous or next note without using the mouse |
| Alt + Shift + Left / Right | Expand Selection | Adds the previous or next note to your current selection |
| X | Flip Stem | Flips note stems up or down |
| 1 .. 6 | Pick Note Length | Sets note value for newly drawn notes (1=Whole .. 6=32nd) |

| | | |
|---|---|---|
| C | Convert Selected Notes | Converts selected notes to the active note length (1..6) |
| Ctrl + Up / Down | Move to Staff Above / Below | Moves selected notes to the adjacent displayed staff |
| Ctrl + 1 .. 7 | Pick Grid Size | Ctrl+7=Whole, Ctrl+6=Half, Ctrl+5=Quarter, Ctrl+4=8th, Ctrl+3=16th, Ctrl+2=32nd, Ctrl+1=Fre |
| Ctrl + K / [Link] Button | Link Grid & Note | Keeps your Grid size and Note Length matched automatically |
| . (Period) | Dotted Note | Turns a dotted note (1.5x length) on or off |
| - / 0 / = | Sharps & Flats | -=Flat (b), 0=Natural, ==Sharp (#) |
| S | Slur (Legato) | Adds or removes a smooth legato slur across notes |
| T | Tie (Hold Note) | Ties two notes of the same pitch together |
| A | Accent | Adds or removes an accent mark (>) on selected notes |
| Q | Quantize Window | Opens the Quantize timing dialog |
| Ctrl + L | Edit Lyrics | Opens the lyric text box on the selected note |
| Ctrl + T | New Text Note | Places a movable text label at the cursor |
| Ctrl + S | Save Project | Saves your REAPER project |
| Shift + Mouse Wheel | Scroll Left / Right | Scrolls horizontally across the bars |
| Middle Mouse Drag | Pan Page | Drags the score page around like a hand tool |
| Ctrl + Mouse Wheel | Zoom In / Out | Makes the sheet music larger or smaller |

# Chapter 18: How REAPER-Notator Keeps Your Sheet Music Looking Professional

## 18.1 Automatic Bar Widths & Natural Note Spacing

You never have to tug individual bars wider by hand so your notes fit:

- Balanced Spacing: Whole and half notes take up a comfortable amount of room without wasting space, while fast 16th and 32nd notes are given plenty of breathing room.
- Busy Bars Automatically Expand: Whenever a bar has lots of fast notes, chords, sharps, flats, or dots, REAPER-Notator automatically widens that bar so symbols never crash into each other.

## 18.2 Automatic Stem Directions & Stem Lengths

How Stems Are Positioned Automatically:

- Middle Line Rule: Notes below the middle line have stems pointing up; notes on or above the middle line have stems pointing down (unless they are part of a beamed melody group).
- High & Low Ledger Line Notes: Notes sitting far above or below the staff automatically get longer stems that reach all the way back to the middle line of the staff.
- Fast Notes With Multiple Flags: Very fast 32nd and 64th notes automatically get slightly taller stems so all their flags stay neat and clear of the notehead.

## 18.3 Clear Beaming Across Beats & Rests

How Beams Help You Read Rhythm at a Glance:

- Grouped by the Beat: Beams break naturally between main beats so you can always see where Beat 1, 2, 3, and 4 fall in any time signature (including 6/8, 5/8, and 7/8).
- Automatic Beam Breaks at Rests: Beams split cleanly around rests so a beam line never cuts through a rest symbol.
- Gentle Slants: Beam angles follow the rise and fall of your melody without tilting too steeply.

## 18.4 Neat Chords & Dotted Notes

- Neighbouring Notes in a Chord: When two notes in a chord sit on adjacent line-and-space steps, they automatically sit on opposite sides of the stem so both noteheads are clearly visible.
- Dots Never Sit on a Staff Line: Rhythm dots on a line note are automatically placed in the clear space just above the line so they never get hidden by the staff line.

## 18.5 Smart Sharps, Flats & Naturals Within Each Bar

- No Cluttered Repeat Signs: Once a sharp or flat appears on a note in a bar, repeated notes on that same line or space in the same bar don't repeat the accidental unnecessarily.
- Automatic Natural Cancellation Signs: If a note was sharp earlier in the bar and returns to its natural pitch later in the same bar, REAPER-Notator automatically displays a natural sign for you.
- Stacked Chord Accidentals: When a big chord has several sharps or flats at once, they automatically stagger into neat diagonal columns so they never overlap.

## 18.6 Clean Placement of Staccato Dots, Accents & Fermatas

- Staccato & Tenuto Marks Avoid Staff Lines: Dots and tenuto bars are placed in clear spaces so staff lines never cut through them.
- Accents & Fermatas Stay Clear of the Staff: Accents, marcatos, and fermatas automatically sit outside the staff lines and clear of note stems.

## 18.7 Graceful Ties & Slurs

- Classic Tapered Curves: Ties and slurs are drawn with a traditional engraved crescent shape-thin at the tips and slightly fuller in the middle.
- Chord Ties Curve Outward: When you tie a full chord, the upper ties curve gently upward and the lower ties curve downward.
- Slurs Arch Cleanly Over Stems & Dots: Phrase slurs automatically arch outside of staccato dots, ties, and stem tips.

## 18.8 Rests, Triplets/Tuplets, Glissandos & Piano Braces

- Clean Triplets & Tuplets: Triplet and tuplet brackets automatically adjust their height to clear all noteheads, stems, and articulations inside the group-including triplets that contain rests.
- Glissando Lines Leave Room for Dots & Sharps: Wavy glissando and straight portamento lines stop neatly before any sharp or flat on the target note.
- Automatic Brackets & Curved Piano Braces: Instrument families are grouped with straight orchestral brackets, while Grand Staff instruments (like Piano, Harp, and Celesta) automatically receive a classic curved brace on the left.

## 18.9 Centered Tremolo Slashes & Rock-Solid Reliability

- Tremolo Slashes: Single, double, and triple tremolo slashes are centered right on the note stem at every zoom level.
- Built-In Crash Protection & 209 Automated Quality Checks: Every window, panel, and musical tool in REAPER-Notator is continuously verified by 209 automated quality tests so your score editing and playback stay rock-solid.

## 18.10 Tempo-Mapped Audio Wave Tracks & Audio-to-Notation Suite

- Separate Visibility Switch (Default Always OFF): Audio Wave Tracks are strictly disabled by default on every startup so standard MIDI scores remain clean. Toggle 'Audio Wave Tracks (Tempo-Mapped)' anytime in the Top Bar View menu or Settings -> General Settings.
- Dedicated [AUDIO] Track Badge: Audio tracks display a prominent teal [AUDIO] badge in both regular and sticky track headers, alongside interactive [Wave / Notes] and [To MIDI] action pills.
- Exact REAPER Tempo-Map Synchronization & 2D Viewport Culling: Audio waveforms stretch and compress sub-pixel-accurately across BPM changes, linear tempo ramps, and dynamic measure widths, while strict 2D viewport culling ensures off-screen tracks and off-screen audio slices are never computed.
- Configurable Waveform Resolution, Custom Color & Invert Mode: Choose Low (6 px), Medium (3 px), High (2 px), or Ultra (1 px) resolution in Settings -> General Settings, customize the Audio Waveform color (or use the REAPER track color) in Settings -> Appearance & Colors, and enjoy automatic high-contrast adaptation in Invert Mode (Dark Theme).
- Read-Only Notation Preview & Copy Audio Notes to New MIDI Track: Switch any audio track between its tempo-mapped Waveform View and Read-Only Notation Preview, or click 'Audio -> MIDI' in the toolbar to copy transcribed notes onto a newly created, fully editable REAPER MIDI track.